Intangible Capital and the Technology of Scale : Returns to Scale, Markups, and Knowledge Spillovers

Yusuf Ozkara*

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Abstract

This paper examines how intangible capital reshapes scale economies, markups, and knowledge diffusion using a structural nonparametric production framework that incorporates intangibles as a factor input and external R&D as a productivity shifter. Using U.S. public firm data from 1975 to 2024, I document that returns to scale increase sharply with firm size only when intangible capital is included — absent intangibles, returns to scale are approximately constant across the size distribution. Markups rise with intangible intensity but follow an inverted-U shape in firm size, indicating that pricing power reflects asset composition rather than scale per se. Incorporating intangibles significantly lowers estimated aggregate markups, revealing that traditional models overstate market power by misattributing technological scale effects to price-cost margins. R&D spillover exposure has declined over time, as large incumbent firms increasingly internalize innovation. Productivity declines with firm size across the entire intangible distribution, while intangible-intensive firms exhibit higher relative productivity. These results suggest that the rise of superstar firms reflects a structural transformation toward a production regime where scalable intangible assets—rather than traditional productivity shocks — govern firm growth and market structure.

JEL Codes: D22, D24, E22, L20

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*T-Mobile USA, Inc. Email: yusufozkara.usa@gmail.com. The views expressed are my own and do not reflect those of T-Mobile.

1 Introduction

Over the last several decades, advanced economies have undergone a fundamental transformation. Industry concentration has risen (Autor et al. (2020); Van Reenen (2018)), markups have increased in many sectors (De Loecker et al. (2020)), and productivity dispersion between frontier and laggard firms has widened (Akcigit and Ates (2023)). At the same time, aggregate productivity growth has slowed and knowledge diffusion across firms has weakened (Goldin et al. (2024)). While existing explanations typically attribute these trends to either superior firm selection or rising market power, a deeper question remains: has the underlying production technology itself changed in ways that structurally favor scale?

This paper argues that the rise of the modern "superstar" firms reflects a shift in production functions driven by the accumulation of intangible capital. While prior work emphasizes the macroeconomic importance of intangible investment and its role in firm valuation and growth (Corrado et al. (2009); Crouzet and Eberly (2020)), less is known about how intangible capital reshapes the production function itself. Are increasing returns a mechanical byproduct of firm size, or do they arise from differences in asset composition? Do large firms grow because they are more productive, or because intangible assets enable scalable expansion? And how does the accumulation of intangible capital affect knowledge diffusion across firms?

To address these questions, it is important to understand how intangible capital differs from physical capital. Intangible assets—such as patents, software, proprietary data, and organizational know-how—possess two defining economic characteristics: non-rivalry in use and limited excludability (Crouzet et al. (2022)).

First, intangible capital is largely non-rival within the firm. Because it represents codified or tacit information, it can be deployed simultaneously across multiple production processes without being depleted. A software algorithm, for example, can coordinate thousands of machines at once, whereas each physical machine performs only a single task at a time. This feature implies that the marginal cost of scaling production declines with firm size, generating increasing returns for firms able to deploy intangible assets at scale.

Second, intangible capital—particularly knowledge capital—exhibits limited excludability[1]. Although patents and intellectual property rights provide some protection, knowledge can often be imitated, reverse-engineered, or used to inspire related innovations. This partial non-excludability generates knowledge spillovers across firms (Bloom et al. (2013)).

[1]Unlike knowledge capital, organizational capital (such as firm-specific management practices, internal processes, workforce expertise, and distribution systems) is difficult to codify and transfer (Lev and Radhakrishnan (2003); Lev et al. (2016).). It is therefore largely firm-specific and does not generate comparable spillovers.

Together, these features suggest a production regime in which intangible accumulation may simultaneously amplify scale economies within firms while weakening economy-wide knowledge diffusion. To study these forces, I develop a structural nonparametric production framework that incorporates intangible capital as a distinct factor input and external R&D as a productivity shifter. I estimate nonparametric production functions following the methodology of Demirer (2025), allowing output elasticities, returns to scale, markups, and spillovers to vary flexibly across firms while incorporating both labor-augmenting and Hicks-neutral productivity [2]. This flexibility is important because if intangibles alter the curvature of the production function heterogeneously across firms, imposing a parametric Cobb–Douglas structure will not capture the mechanism.

Using panel data on U.S. public firms from 1975 to 2024, I estimate firm-level production functions and recover measures of scale economies, market power, productivity, and knowledge spillovers. I measure intangible capital as the sum of accumulated R&D and the organizational capital following Peters and Taylor (2017) and Ewens et al. (2025). The long time span captures the transition from a predominantly physical-capital-based production regime to one increasingly driven by intangible assets.

The first and central finding is that returns to scale increase sharply with firm size only when intangible capital is included in the production function. When intangibles are excluded, estimated returns to scale remain approximately constant across the firm-size distribution. Once intangible capital is incorporated, however, returns to scale rise monotonically with both firm size and intangible intensity. Among large, intangible-intensive firms, estimated returns to scale exceed 1.6, whereas small and low-intangible firms operate close to constant returns.

This result has two key implications. First, increasing returns are not simply a mechanical consequence of physical scale. Instead, scale economies arise primarily through the accumulation of intangible assets. Second, scalability is highly heterogeneous: firms of similar physical size exhibit markedly different returns depending on their intangible intensity. These findings provide a production-based mechanism for the rise of large firms documented by Autor et al. (2020). Rather than reflecting purely superior productivity, firm dominance may emerge from scalable intangible assets that allow output to expand disproportionately relative to inputs.

The rise in scale economies is driven by a systematic reallocation of factor elasticities. Labor elasticity declines with firm size but increases with intangible intensity, whereas physical

[2] Distinguishing between Hicks-neutral and labor-augmenting productivity is important because a large literature argues that technological change—particularly skill-biased technological change—is closely linked to the diffusion of intangible assets and knowledge capital (e.g., Krusell et al. (2000); Acemoglu (2002))).

capital elasticity rises with size and declines with intangible intensity. In contrast, materials elasticity remains largely flat across both size and intangible intensity distributions. Together, these patterns point to a structural shift toward intangible–labor complementarity in modern production.

Consistent with the broader macroeconomic evidence on declining labor shares, I also document a secular decline in both labor and physical capital elasticities over time (Autor et al. (2020)), accompanied by a substantial increase in intangible elasticity. These patterns suggest that the growing dominance of large firms reflects not only changes in market structure but a deeper transformation in production technology, in which scalable intangible assets reshape the technology of scale.

Second, markups increase monotonically with intangible intensity but exhibit an inverted-U relationship with firm size [3]. This non-monotonic pattern suggests that pricing power is driven more by asset composition than by scale per se, and that scale and market power are not synonymous. Markups rise from small to mid-sized firms but decline among the largest firms, indicating that large firms do not necessarily charge the highest margins. One interpretation is a volume–margin trade-off. Mid-sized firms may leverage intangible assets to command higher markups in specialized product niches, whereas the largest firms prioritize market share. Firms with high returns to scale can expand output at relatively low marginal cost and may therefore set lower markups while earning high profits through volume. This behavior is consistent with models of strategic pricing and market-share expansion (Milgrom and Roberts (1982); Atkeson and Burstein (2008)) and with recent work showing that highly productive firms may pass through cost advantages to expand sales (Edmond et al. (2023); Baqaee and Farhi (2020)).

This result also refines the evidence on rising markups documented by De Loecker et al. (2020). While markups increase with intangible intensity, I show that incorporating intangible capital into the production function lowers aggregate markup estimates [4]. When intangible capital is omitted, part of the output response attributable to scalable intangible inputs is incorrectly attributed to flexible inputs, inflating implied markups. Traditional production-function estimates therefore overstate market power by misattributing technological scale effects to price–cost margins (Bond et al. (2021); Koh et al. (2020)).

Third, R&D spillover exposure declines with firm size but increases with intangible intensity. Firms with greater intangible intensity appear to benefit more from external knowledge, consistent with complementarities between internal capabilities and external ideas (Bloom

[3] Intangible intensity is defined as the share of intangible capital in firm's total capital stock (the sum of intangible and physical capital).

[4] Following De Loecker et al. (2020), aggregate markups are computed as the sales-weighted averages of firm-level markups within each year.

et al. (2013); Akcigit and Ates (2023)). At the same time, larger firms exhibit substantially lower exposure to spillovers from other firms. As economic activity shifts toward these large firms, sales-weighted spillover exposure declines over time.

These patterns are consistent with increased internalization of innovation among dominant firms. Firms with large intangible stocks may rely more heavily on internally generated knowledge and therefore depend less on external spillovers. This evidence connects to the literature on innovation networks and declining knowledge diffusion (Akcigit and Ates (2023)) and suggests a production-based mechanism: scalable intangible assets enable firms to expand internally while reducing reliance on external knowledge flows. As scale rises through intangible accumulation, the marginal contribution of external knowledge to productivity declines.

Finally, after controlling for industry-year fixed effects, Hicks-neutral productivity declines with firm size across the entire intangible distribution, while intangible-intensive firms exhibit higher relative productivity. These findings complement the evidence on productivity dispersion documented by Syverson (2011) and the misallocation literature (Hsieh and Klenow (2009)), but suggest a more nuanced interpretation of firm growth. Firms may expand even when their relative productivity declines because intangible capital generates scale economies that allow output to grow disproportionately relative to inputs. Large firms are therefore not necessarily more productive in the Hicks-neutral sense; rather, they are more scalable.

This paper contributes to several strands of the literature. First, it contributes to the literature on market power by showing that markups are closely related to firms' intangible intensity rather than firm size alone (De Loecker et al. (2020)). Second, it provides a technological mechanism for the rise of superstar firms (Autor et al. (2020)) by identifying scalable intangible capital as a key source of increasing returns. Third, it connects to the misallocation and productivity dispersion literature (Hsieh and Klenow (2009); Syverson (2011)) by documenting declining within-industry productivity with firm size alongside strong scale incentives. Finally, it contributes to the literature on innovation and knowledge diffusion (Bloom et al. (2013); Akcigit and Ates (2023)) by showing that spillover exposure declines as economic activity becomes concentrated among large firms. More broadly, the findings suggest that modern production has shifted toward a regime in which scalable intangible assets govern firm growth, market structure, and knowledge diffusion.

This paper is organized as follows. Section 2 presents the model and the assumptions required for estimation. Section 3 describes the identification strategy for output elasticities, markups, and knowledge spillovers. Section 4 introduces the data and estimation strategy. Section 5 presents the main empirical results. Section 6 reports robustness checks. Section

7 concludes.

2 Model

This section develops a nonparametric production function framework that accommodates both intangible capital and knowledge spillovers. I extend the framework of Demirer (2025) by introducing intangible capital as a distinct factor input and allowing firm productivity to depend on external knowledge through R&D spillovers. After outlining the production technology and the firm's timing assumptions, I show how flexible input ratios and intermediate material demand can be exploited to recover labor-augmenting and Hicks-neutral productivity. Finally, I construct the control variables required to identify factor elasticities, markups, and spillover effects.

2.1 Production Function

In each period t, firm i produces output Y_{it} according to the following production technology:

$$Y_{it} = F_t(K_{it}, K_{it}^{int}, \omega_{it}^L L_{it}, M_{it}) exp(\omega_{it}^H) exp(\epsilon_{it}) \tag{1}$$

where K_{it} denotes physical capital, K_{it}^{int} is intangible capital, M_{it} represents intermediate materials, and L_{it} is labor. The model accommodates multi-dimensional unobserved heterogeneity. Specifically, $\omega_{it}^L \in \mathbb{R}$ captures labor-augmenting productivity, such that $\omega_{it}^L L_{it}$ represents effective labor. Furthermore, $\omega_{it}^H \in \mathbb{R}$ captures Hicks-neutral productivity, which symmetrically scales output for any given input bundle. Finally, $\epsilon_{it} \in \mathbb{R}$ represents a random idiosyncratic shock to production.

The firm's inputs are partitioned into two groups according to their dynamic properties. Physical and intangible capital are *predetermined*: firms choose investment in period $t-1$, fixing the capital stocks before the realization of productivity. In contrast, labor and materials are *flexible* inputs. Firms choose flexible inputs in period t to minimize production costs, conditional on their current information set, $\mathcal{I}_{it}$. This information set includes both productivity components $(\omega_{it}^L, \omega_{it}^H)$, current capital stocks, and the firm's past history. By assumption, the idiosyncratic shock is orthogonal to the firm's information set, $\mathbb{E}[\epsilon_{it} \mid \mathcal{I}_{it}] = 0$, implying that it is unobserved by the firm when making input decisions.

Input markets are assumed to be perfectly competitive, implying that firms take input prices as given. In contrast, the output market is characterized by imperfect competition, allowing firms to exert market power and charge heterogeneous markups over marginal costs.

A distinctive feature of the model is the incorporation of intangible capital into the production function. Including intangible assets—such as intellectual property, software, and organizational capital—is essential for capturing the modern economy's reliance on non-physical capital accumulation (Corrado et al. (2005)). This role of intangible assets has become even more pronounced with recent technological advances. As Crouzet and Eberly (2019) document, both the intensity and the impact of intangible capital vary substantially across industries and over time. To accommodate this variation, as well as broader macroeconomic shifts such as the decline in the labor share (Autor et al. (2020)), I allow the production function $F_t(\cdot)$ to be industry-specific and time-varying.

Industry and time are not the only sources of heterogeneity in production; substantial *cross-firm* heterogeneity also exists. For example, large firms tend to be more capital intensive, whereas smaller firms exhibit relatively greater labor intensity (Holmes and Schmitz (2010)). The nonparametric production function framework accommodates this heterogeneity by allowing factor inputs to flexibly determine firm-level production outcomes.

To manage the dimensionality of unobserved productivity, the empirical strategy permits unobserved heterogeneity in only one flexible input factor. I specify this as labor-augmenting productivity for three distinct reasons. First, this choice aligns with the framework of Doraszelski and Jaumandreu (2018), who demonstrate that technological change is predominantly labor-augmenting. Second, cross-sectional variation in labor costs substantially exceeds that of other inputs, suggesting significant unobserved heterogeneity in workforce composition and efficiency. Finally, intangible capital inherently relies on non-excludable knowledge. Because workers frequently internalize and transfer the fruits of R&D across firm boundaries, the external spillover effects of intangible assets manifest most naturally through improvements in labor productivity.

2.2 Assumptions

This section details the structural assumptions necessary to identify and estimate the firm's production technology. The foundational assumption is weak homothetic separability, which permits the expression of labor productivity as a function of observed input choices. Building on Levinsohn and Petrin (2003), I then establish conditions to invert Hicks-neutral productivity from intermediate input demand. Finally, I define the evolution of both productivity processes, explicitly formalizing the spillover effects of external knowledge capital.

Assumption 2.1: Weak Homothetic Separability

I assume the production technology takes the functional form:

$$Y_{it} = F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it}))exp(\omega_{it}^H)exp(\epsilon_{it}) \tag{2}$$

where $h_t(\cdot)$ is a homogeneous function. This implies that the production function is weakly separable, partitioning the predetermined inputs (physical and intangible capital) from a homogeneous function of flexible inputs (materials and effective labor). While this imposes structure on the production technology, this assumption nests widely used functional forms. For instance, the Cobb-Douglas production function:

$$Y_{it} = K_{it}^{\alpha}(K_{it}^{int})^{\beta}(\omega_{it}^{L}L_{it})^{\gamma}M_{it}^{\theta}exp(\omega_{it}^{H})exp(\epsilon_{it})$$

satisfies this condition, as $h(\cdot) = (\omega_{it}^{L}L_{it})^{\gamma}M_{it}^{\theta}$ is homogeneous and separable from the capital inputs. However, Cobb-Douglas rigidly imposes a constant unit elasticity of substitution and fixed output elasticities across firms.

Similarly, the Constant Elasticity of Substitution (CES) functional form:

$$Y_{it} = \left(\beta_k K_{it}^{\sigma} + \beta_I (K_{it}^{int})^{\sigma} + \beta_l (\omega_{it}^{L}L_{it})^{\sigma} + (1-\beta_k-\beta_I-\beta_l)M_{it}^{\sigma}\right)^{\upsilon/\sigma} exp(\omega_{it}^{H})exp(\epsilon_{it})$$

satisfies Assumption 2.1, with $h(\cdot) = \beta_l(\omega_{it}^{L}L_{it})^{\sigma} + (1-\beta_k-\beta_I-\beta_l)M_{it}^{\sigma}$. While CES is less restrictive than Cobb-Douglas, it still imposes a constant elasticity of substitution, σ, across all firms. By relying solely on weak homothetic separability, the empirical framework maintains a highly flexible, nonparametric form that accommodates heterogeneous output elasticities and non-constant substitution patterns across the firm distribution.

Assumption 2.2: Firms minimize their costs

Firms choose flexible inputs (labor and materials) in period t to minimize short-run costs, conditional on their predetermined capital stocks. This static optimization framework also facilitates the recovery of firm-level markups from the first-order conditions, following the methodology of Loecker and Warzynski (2012).

Firms solve the following cost minimization problem given competitive input prices and output demand, $\tilde{Y}_{it}$:

$$\min_{L_{it},M_{it}} \left\{ p_t^L L_{it} + p_t^M M_{it} : F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^{L}L_{it}, M_{it}))exp(\omega_{it}^{H})exp(\epsilon_{it}) \geq \tilde{Y}_{it} \right\}$$

where p_t^L and p_t^M are the respective prices of labor and materials at time t. Because input markets are perfectly competitive, these prices vary over time but are common across firms.

By Shepherd's lemma and Assumption 2.1, this cost minimization yields the optimal material-to-labor ratio (proof in Appendix):

$$\tilde{M}_{it} = \frac{M_{it}}{L_{it}} = r_t(\omega_{it}^{L}) \tag{3}$$

where $r_t(\cdot)$ is an unknown function. Crucially, Equation 3 establishes that the optimal flexible input ratio is determined entirely by labor-augmenting productivity and is independent of Hicks-neutral productivity.

Assumption 2.3: Elasticity of substitution between effective labor and material is less than 1 or greater than 1.
To invert $r_t(\cdot)$ and recover unobserved labor-augmenting productivity, the material-to-labor ratio must be strictly monotonic in ω_{it}^L. The elasticity of $\tilde{M}_{it}$ with respect to ω_{it}^L is strictly given by $1 - \sigma_t(\omega_{it}^L L_{it}, M_{it})$, where $\sigma_t(\cdot)$ is the elasticity of substitution between effective labor and materials. I assume $\sigma_t(\cdot) \neq 1$. Consequently, $\tilde{M}_{it}$ is strictly increasing or strictly decreasing in ω_{it}^L. Under this condition, ω_{it}^L can be expressed as an inverse function of the flexible input ratio:

$$\omega_{it}^L = r_t^{-1}(\tilde{M}_{it}) \tag{4}$$

Unobserved labor productivity can now be characterized as an unknown function of the material-to-input ratio.

Assumption 2.4: Material demand is monotone in Hicks-neutral productivity
Following Levinsohn and Petrin (2003), I assume material demand is strictly increasing in Hicks-neutral productivity, conditional on capital and labor-augmenting productivity:

$$M_{it} = s_t(K_{it}, K_{it}^{int}, \omega_{it}^L, \omega_{it}^H) \tag{5}$$

where $s_t(.)$ is a strictly increasing function in ω_{it}^H. In other words, more productive firms demand higher materials. This framework extends the standard Levinsohn-Petrin approach in two dimensions: it incorporates labor-augmenting productivity alongside Hicks-neutral productivity, and it explicitly includes both physical and intangible capital in the state space.

By strict monotonicity, ω_{it}^H can be inverted as a function of factor inputs. Substituting Equation 4 into this inversion isolates Hicks-neutral productivity from ω_{it}^L:

$$\omega_{it}^H = s_t^{-1}(K_{it}, K_{it}^{int}, r_t^{-1}(\tilde{M}_{it}), M_{it}) = \bar{s}_t(K_{it}, K_{it}^{int}, \tilde{M}_{it}, M_{it}) \tag{6}$$

where $\bar{s}_t(.)$ is an unknown function. ω_{it}^H becomes a function of factor inputs, only determined by physical capital, intangible capital, material-to-labor ratio and materials. As a result, it is observable by the econometrician.

Assumption 2.5: Physical and intangible capital are predetermined variables
The physical and intangible capital are determined by investments in the preceding period,

capturing a time-to-build dynamic (Kydland and Prescott (1982)):

$$K_{it} = \kappa_{1t}(K_{it-1}, I_{it-1})$$
$$K_{it}^{int} = \kappa_{2t}(K_{it-1}^{int}, I_{it-1}^{int})$$

where I_{it} and I_{it}^{int} are physical and intangible investments, respectively.The unknown functions $\kappa_{1t}(\cdot)$ and $\kappa_{2t}(\cdot)$ accommodate unobserved fixed and convex adjustment costs. This timing assumption implies that firms start to fully realize the benefits of physical and intangible investments within a year in their production processes.

Assumption 2.6: Productivity shocks have a first-order Markov process The unobserved productivity shocks follow a joint first-order Markov process, explicitly augmented to incorporate external knowledge spillovers:

$$P(\omega_{it}^L, \omega_{it}^H \,\Big|\, \mathcal{I}_{t-1}) = P(\omega_{it}^L, \omega_{it}^H \,\Big|\, \omega_{it-1}^L, \omega_{it-1}^H, \sum_{j \in I, j \neq i} R_{jt-1})$$

where R_{jt} is the knowledge capital stock of company j at time t.This specification allows firm i i's productivity to depend not only on its own past productivity but also on the accumulated knowledge capital of other firms. Because knowledge capital is characterized by limited excludability, innovations generated by one firm naturally create positive externalities for others operating in similar technological or industry environments (Bernstein and Nadiri (1988); Bloom et al. (2013)). Consequently, exposure to a larger pool of external R&D shifts the conditional distribution of a firm's future productivity upward, formally capturing the mechanics of intra-industry knowledge spillovers.

Assuming the productivity processes are continuous, I utilize the Skorohod representation of random variables to define labor-augmenting productivity as:

$$\omega_{it}^L = g_1(\omega_{it-1}^L, \omega_{it}^H, \sum_{j \in I, j \neq i} R_{jt-1}, u_{it}^1), \quad u_{it}^1 \mid \omega_{it-1}^L, \omega_{it}^H, \sum_{j \in I, j \neq i} R_{jt-1} \sim \text{Uniform}(0,1)$$

u_{it}^1 represents an unobserved innovation term. Unlike standard approaches, this disturbance is non-separable. Intuitively, u_{it}^1 captures a firm's idiosyncratic labor productivity rank, conditional on its historical productivity and the aggregate R&D spillovers it absorbs.

Similarly, Hicks-neutral productivity is defined as:

$$\omega_{it}^H = g_2(\omega_{it-1}^L, \omega_{it-1}^H, \sum_{j \in I, j \neq i} R_{jt-1}, u_{it}^1, u_{it}^2), \quad u_{it}^2 \mid \omega_{it-1}^L, \omega_{it}^H, \sum_{j \in I, j \neq i} R_{jt-1}, u_{it}^1 \sim \text{Uniform}(0,1)$$

where u_{it}^2 represents the conditional Hicks-neutral productivity rank after accounting for the

past productivity and the aggregate R&D spillovers.

3 Estimation Strategy: A Control Variable Approach

To resolve the endogeneity of flexible inputs, I use the control function approach of Imbens and Newey (2009). This methodology is uniquely suited for models with non-separable, multidimensional disturbances, whereas standard proxy variable techniques (Olley and Pakes (1996)) restrictively assume a single, additively separable scalar unobservable.

A control variable is characterized by the conditional distribution function of the endogenous variable conditioning on the instruments. The control variables have two key characteristics: they are strictly monotone in the endogenous variable and independent from the instruments. I construct distinct control variables for both labor-augmenting and Hicks-neutral productivity.

Let $W_{it-1} = (K_{it-1}, K^{int}_{it-1}, \tilde{M}_{it-1}, M_{it-1})$ denote the vector of lagged inputs. By substituting the labor productivity process into the flexible input ratio derived in Equation 3, $\tilde{M}_{it}$ can be written as:

$$\begin{aligned}
\tilde{M}_{it} &= r_t\big(\omega^L_{it}\big) = r_t\Big(g_1\big(\omega^L_{it-1}, \omega^H_{it}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}\big)\Big) \\
&= r_t\Big(g_1\Big(\bar{r}_t\big(\tilde{M}_{it-1}\big), \bar{s}_t(K_{it-1}, K^{int}_{it-1}, \tilde{M}_{it-1}, M_{it-1}), \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}\Big)\Big) \\
&= \tilde{r}_t\Big(W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}\Big)
\end{aligned}$$

where $\tilde{r}_t(.)$ is an unknown function. Because ω^L_{it} is strictly monotonic in u^1_{it} (by the Skorohod representation) and $\tilde{M}_{it}$ is strictly monotonic in ω^L_{it} (Assumption 2.3), $\tilde{M}_{it}$ is strictly monotonic in u^1_{it}. Furthermore, the timing assumptions dictate that u^1_{it} is independent of the predetermined inputs W_{it-1} and aggregate knowledge spillovers (proof in Appendix).

Because u^1_{it} satisfies both the monotonicity and independence criteria and is uniformly distributed by definition, it can be recovered as the conditional cumulative distribution function (CDF) of the flexible input ratio:

$$u^1_{it} = F_{\tilde{M}_{it}|W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}}\Big(\tilde{M}_{it} \mid W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}\Big) \tag{7}$$

where $F_{\tilde{M}_{it}|W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}}(.)$ is the cumulative distribution function of $\tilde{M}_{it}$ conditional on $W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}$. That means that if two firms share identical historical input trajectories and are exposed to the exact same volume of knowledge spillovers, the firm exhibiting

the higher material-to-labor ratio must possess a higher productivity rank, u^1_{it}. Having isolated u^1_{it}, labor-augmenting productivity is now fully characterized by observable data and the recovered control variable:

$$\begin{aligned}
\omega^L_{it} &= g_1\Big(\omega^H_{it-1}, \omega^L_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}\Big) \\
&= g_1\Big(\bar{r}_t\big(\tilde{M}_{it-1}\big), \bar{s}_t(K_{it-1}, K^{int}_{it-1}, \tilde{M}_{it-1}, M_{it-1}), \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}\Big) \\
&= c_{1t}\Big(W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}\Big) \qquad (8)
\end{aligned}$$

where $c_{1t}(.)$ is an unknown function. One can, similarly, define the control variable for the Hicks-neutral productivity using the monotonicity assumption,

$$\begin{aligned}
M_{it} &= s_t(K_{it}, K^{int}_{it}, \omega^L_{it}, \omega^H_{it}) \\
&= s_t\Big(K_{it}, K^{int}_{it}, g_1\big(\omega^L_{it-1}, \omega^H_{it}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}\big), g_2\big(\omega^L_{it-1}, \omega^H_{it}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}, u^2_{it}\big)\Big)
\end{aligned}$$

Substituting equation 4 and 6 into ω^L_{it} and ω^H_{it} in the equation above, material demand becomes

$$M_{it} = \tilde{s}_t\big(K_{it}, K^{int}_{it}, W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}, u^2_{it}\big)$$

where $\tilde{s}_t()$ is an unknown function. The timing assumptions guarantee that the conditional Hicks-neutral innovation, u^2_{it}, is independent of predetermined capital, the lagged state variables W_{it-1}, and the labor-augmenting innovation u^1_{it}. Furthermore, material demand is strictly monotonically increasing in u^2_{it}. Consequently, the control variable for Hicks-neutral productivity is identified via the conditional CDF of material demand:

$$u^2_{it} = F_{M_{it}|K_{it}, K^{int}_{it}, W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}}\Big(M_{it} \mid K_{it}, K^{int}_{it}, W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}\Big) \qquad (9)$$

where $F(.)$ is the CDF of M_{it} conditional on $K_{it}, K^{int}_{it}, W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}$. This implies that conditional on capital stocks, historical inputs, and knowledge spillovers, differences in material demand strictly identify variations in the Hicks-neutral productivity rank. Finally, ω^H_{it} can be nonparametrically expressed as:

$$\omega^H_{it} = c_{2t}\big(W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}, u^2_{it}\big) \qquad (10)$$

where $c_{2t}(.)$ is an unknown function. This indicates that, conditional on the two control variables, external knowledge spillovers, and past factor inputs, there is no remaining residual variation in Hicks-neutral productivity.

4 Identification

This section outlines the identification of the output elasticities of factor inputs, firm-level markups, and the productivity effects of knowledge spillovers.

The cost minimization problem for the flexible inputs (materials and labor) yields the following optimality condition (proof in Appendix):

$$\frac{\theta_{it}^M}{\theta_{it}^L} = \frac{\alpha_{it}^M}{\alpha_{it}^L} \tag{11}$$

where θ_{it}^M and θ_{it}^L denote the output elasticities of materials and labor, respectively, and α_{it}^M and α_{it}^L represent their corresponding revenue shares. This condition establishes that the ratio of the flexible input elasticities strictly equals the ratio of their observed revenue shares. I exploit this proportionality to point-identify the individual elasticities.

Let lower-case letters denote logarithms. The logarithmic production function is given by:

$$y_{it} = f_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it})) + \omega_{it}^H + \epsilon_{it}$$

where $f_t(\cdot)$ is the logarithm of the $F_t(\cdot)$ function. Assuming the homogeneous function $h_t(\cdot)$ is homogeneous of degree one, effective labor can be factored out: $h_t(\omega_{it}^L L_{it}, M_{it}) = L_{it} h_t(\omega_{it}^L, \tilde{M}_{it})$. Substituting the inverted labor productivity from Equation 4, this aggregate flexible input function becomes $L_{it} h_t(r_t(\tilde{M}_{it}), \tilde{M}_{it}) \equiv L_{it} \bar{h}_t(\tilde{M}_{it})$, where $\bar{h}_t(\cdot)$ is an unknown function.

Because both arguments of the original $h_t(\cdot)$ function depend on $\tilde{M}_{it}$, fully identifying its internal structure is infeasible. However, identifying the composite function $\bar{h}_t(\cdot)$ is sufficient to recover the output elasticities. The production function can thus be rewritten as:

$$y_{it} = f_t(K_{it}, K_{it}^{int}, L_{it} \bar{h}_t(\tilde{M}_{it})) + \omega_{it}^H + \epsilon_{it}$$

Building upon this framework, the following proposition establishes the identification of the flexible input elasticities.

Proposition 1. *The sum of labor and material output elasticities is equal to:*

$$\theta_{it}^M + \theta_{it}^L = f_{t3}(K_{it}, K_{it}^{int}, L_{it} \bar{h}_t(\tilde{M}_{it})) L_{it} \bar{h}_t(\tilde{M}_{it})$$

where $f_{t3}(.)$ is the derivative of $f_t(.)$ with respect to the third argument.

The proof is provided in the appendix. The proposition demonstrates that derivative of $f_t(.)$ with respect to its third argument is equivalent to the sum of flexible input elasticities. Importantly, I do not need to identify $h_t(.)$ function as the sum of elasticities do not depend on the derivative of $h_t()$. Instead, the identification of $\bar{h}_t(.)$ is sufficient to determine the sum of elasticities.

Substituting Equation 11 into this sum of elasticities isolates the individual labor and material elasticities

$$\begin{aligned}\theta_{it}^L &= \frac{\alpha_{it}^L}{\alpha_{it}^L + \alpha_{it}^M} f_{t3}(.) L_{it} \bar{h}_t(\tilde{M}_{it}) \\ \theta_{it}^M &= \frac{\alpha_{it}^M}{\alpha_{it}^L + \alpha_{it}^M} f_{t3}(.) L_{it} \bar{h}_t(\tilde{M}_{it})\end{aligned} \tag{12}$$

Because the input revenue shares, α_{it}, are directly observable in the data, no additional estimation is required to determine them. Once the sum of the elasticities is identified, the individual elasticities are straightforwardly deduced.

Following Loecker and Warzynski (2012) the first order condition of the cost minimization problem yields the expression for the firm-level markup, μ_{it}:

$$\mu_{it} = \frac{\theta_{it}^M}{\alpha_{it}^M} = \frac{\theta_{it}^L}{\alpha_{it}^L} \tag{13}$$

This expression implies that markups can be computed using either the labor elasticity or the material elasticity. Dividing the elasticity of a flexible input by its revenue share yields the firm-level markup. Using the elasticities from Equation 12, I compute markups at the firm level.

I identify the output elasticities of physical and intangible capital using a similar approach.

Proposition 2. *The output elasticity of physical and intangible capital is*

$$\begin{aligned}\theta_{it}^{int} &= f_{t2}(K_{it}, K_{it}^{int}, L_{it}\bar{h}_t(\tilde{M}_{it}))K_{it}^{int} \\ \theta_{it}^K &= f_{t1}(K_{it}, K_{it}^{int}, L_{it}\bar{h}_t(\tilde{M}_{it}))K_{it}\end{aligned} \tag{14}$$

where $f_{t2}(.)$ and $f_{t1}(.)$ is the derivative of $f_t(.)$ with respect to the second and first argument, respectively.

This proposition shows that the output elasticities of physical and intangible capital correspond to the derivatives of the production function with respect to the associated capital

inputs. As before, identifying the composite function $\bar{h}_t(\cdot)$ is sufficient; the internal structure of $h_t(\cdot)$ does not need to be characterized.

Finally, the following proposition identifies the effect of knowledge spillovers on firm productivity.

Proposition 3. *The productivity elasticity of knowledge spillover is*

$$\theta_{it}^{spillover} = \frac{\partial c_{2t}\big(W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}, u_{it}^1, u_{it}^2\big)}{\partial \sum_{j\in I, j\neq i} R_{jt-1}} \frac{\sum_{j\in I, j\neq i} R_{jt-1}}{c_{2t}(.)} \tag{15}$$

The effect of knowledge spillover on firm productivity corresponds to the derivative of the Hicks-neutral productivity function, generated from the control variables, with respect to the sum of knowledge capital.

5 Data and Empirical Model

This section describes the data sources, the construction of key variables, and the empirical strategy used to estimate the model.

5.1 Data

I use firm-level financial statement data from U.S. Compustat to measure intangible capital and estimate the production functions. The sample covers all public U.S. firms from 1975 to 2024. Following standard sampling procedures in the literature, I exclude financial firms (SIC codes 4900–4999), utilities (SIC codes 6000–6999), and government entities (SIC codes 9000 and above). I also restrict the sample to firms with at least 10 employees and drop observations with missing or negative sales, capital, employment, Selling, General, and Administrative (SG&A), or Cost of Goods Sold (COGS). To mitigate the influence of outliers, all those variables are winsorized at the 1% level.

I measure physical capital as the gross value of property, plant, and equipment (Compustat item PPEGT). Labor is defined as the total number of employees, and output is measured as sales. I construct material inputs following the broad methodology of Keller and Yeaple (2009), but with specific modifications to accommodate the intangible capital framework. First, I do not deduct the depreciation rate from total expenses. Second, in addition to subtracting the total wage bill, I explicitly deduct the fraction of SG&A expenditures allocated to organizational capital accumulation. This latter adjustment is crucial to prevent the double-counting of organizational investments as both predetermined intangible capital and flexible material inputs.

Measurement of Intangible Capital. I construct the intangible capital at the firm level as outlined in Ewens et al. (2025) (along with insights from other studies such as Lev and Radhakrishnan (2005), Eisfeldt and Papanikolaou (2014)). Intangible capital consists of two components: knowledge capital and organizational capital.

I measure knowledge capital using reported Research and Development (R&D) expenses. Because R&D investments are recorded as flow variables in Compustat, I accumulate them into a stock variable using the perpetual inventory method:

$$R_{it} = (1 - \delta_{R\&D})R_{it-1} + R\&D_{it} \tag{16}$$

where R_{it} is the knowledge capital of firm i in year t, R&D$_{it}$ is the firm's R&D investment, and $\delta_{R\&D}$ is the industry-specific R&D depreciation rate estimated by Ewens et al. (2025). The initial knowledge stock, R_{i0}, is normalized to zero.

Organizational capital is constructed using Selling, General, and Administrative Expenses (SG&A). Because SG&A encompasses a broad range of overhead and operating expenses, it requires allocating a specific fraction of total SG&A as investment in organizational capital (Lev and Radhakrishnan (2003)). Applying the perpetual inventory method :

$$O_{it} = (1 - \delta_{SG\&A})O_{it-1} + \gamma SG\&A_{it} \tag{17}$$

where O_{it} is the organizational capital stock of firm i in year t, and γ is the industry-specific proportion of SG&A allocated to organizational investment. I use industry-specific estimates of $\delta_{SG\&A}$ and γ from Ewens et al. (2025). The initial organizational stock, O_{i0}, is set to zero.

Total intangible capital stock is therefore the sum of these two components:

$$K_{it}^{int} = R_{it} + O_{it} \tag{18}$$

where K_{it}^{int} is the firm i's intangible capital stock in year t.

All variables constructed are deflated using industry-year price deflators from the Bureau of Economic Analysis (BEA). Annual, industry-level wage data is sourced from the Quarterly Census of Employment and Wages (QCEW) provided by the Bureau of Labor Statistics (BLS).

5.2 Estimation Strategy

I estimate the production function separately across industries and over time. To construct the industry definitions, I broadly categorize firms at the 2-digit NAICS level, aggregating

closely related sectors such as Retail Trade (44–45) and Transportation and Warehousing (48–49). However, to capture nuanced technological and production differences, I explicitly isolate key 3-digit subsectors from their parent categories, breaking out highly specialized industries such as Chemical Manufacturing (325), Computer and Electronic Product Manufacturing (334), and Non-store Retailers (454). Finally, to ensure sufficient statistical power and maintain adequate sample sizes within each industry-year cell, I estimate the model using 8-year rolling windows.

As established in the identification section, the logarithmic production function is given by:

$$y_{it} = f_t(K_{it}, K_{it}^{int}, L_{it}\bar{h}_t(\tilde{M}_{it})) + \omega_{it}^H + \epsilon_{it}$$

Substituting the Hicks-neutral productivity term recovered via the control function approach (Equation 10) yields:

$$y_{it} = f_t(K_{it}, K_{it}^{int}, L_{it}\bar{h}_t(\tilde{M}_{it})) + c_{2t}\big(W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}, u_{it}^1, u_{it}^2\big) + \epsilon_{it} \tag{19}$$

Because the control variables resolve the endogeneity of the flexible inputs, the ex-post shock is strictly orthogonal to the firm's information set: $\mathbb{E}[\epsilon_{it} \mid W_{it}, W_{it-1}, \sum_{j\neq i} R_{jt-1}] = 0$. While this condition permits estimation by minimizing the sum of squared residuals, exploiting the predetermined nature of capital provides additional identifying moments.

To form these moment conditions, I project the productivity process into a separable framework widely utilized in the proxy variable literature (Olley and Pakes (1996)). Specifically, I model Hicks-neutral productivity as:

$$\omega_{it}^H = c_{3t}(\omega_{it-1}^H, \omega_{it-1}^L, \sum_{j\in I, j\neq i} R_{jt-1}) + \upsilon_{it} \tag{20}$$

where $c_{3t}(.)$ is an unknown function, and υ_{it} is separable from $c_{3t}(.)$ with $E[\upsilon_{it}|I_{t-1}] = 0$. This is different from the productivity assumption in the previous section because the innovation terms, u_{it}^1, and u_{it}^2, were non-separable and independent whereas υ_{it} is separable and mean-independent.

Using the lagged state variables, this projection can be rewritten as $\omega_{it}^H = \bar{c}_{3t}(W_{it-1}, \sum_{j\neq i} R_{jt-1}) + \upsilon_{it}$ for an unknown function $\bar{c}_{3t}(\cdot)$. Substituting this into the production function yields the final structural equation:

$$y_{it} = f_t(K_{it}, K_{it}^{int}, L_{it}\bar{h}_t(\tilde{M}_{it})) + \bar{c}_{3t}\big(W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}\big) + \upsilon_{it} + \epsilon_{it} \tag{21}$$

By construction, $\mathbb{E}[\upsilon_{it} + \epsilon_{it} \mid \mathcal{I}_{it-1}] = 0$. Because physical and intangible capital are predetermined, they must be orthogonal to this composite error term, providing the moment conditions $\mathbb{E}[\upsilon_{it} + \epsilon_{it} \mid K_{it}] = 0$ and $\mathbb{E}[\upsilon_{it} + \epsilon_{it} \mid K_{it}^{int}] = 0$.

Combining these conditions with the residual moment from Equation 19, the Generalized Method of Moments (GMM) objective function is:

$$\frac{1}{N}\sum_i \epsilon_{it}^2 + \left(\frac{1}{N}\sum_i (\epsilon_{it} + \upsilon_{it})K_{it}\right)^2 + \left(\frac{1}{N}\sum_i (\epsilon_{it} + \upsilon_{it})K_{it}^{int}\right)^2 \tag{22}$$

Minimizing this objective function requires estimating the control variables and nonparametrically approximating the functions $f_t(\cdot), \bar{h}_t(\cdot), c_{2t}(\cdot)$, and $\bar{c}_{3t}(\cdot)$. The estimation proceeds in steps:

1. **Recovering Control Variables:** I estimate the control variables u_{it}^1 and u_{it}^2 (Equations 7 and 9) using logistic regression. Because the outcome variables are continuous in both cases, I partition their support into a 500-point grid. I estimate the conditional CDFs at these points with second-order polynomials using logistic regression, and then linearly interpolate the remaining points.

2. **Function Approximation:** I approximate the composite flexible input function $\bar{h}_t(\cdot)$ using second-order polynomials. Conditional on $\bar{h}_t(\cdot)$, I approximate the production function $f_t(\cdot)$ and the productivity functions $c_{2t}(\cdot)$ and $\bar{c}_{3t}(\cdot)$ using second-order polynomials.

3. **Optimization and Inference:** I minimize the empirical criterion function (Equation 22) to recover the structural parameters. Output elasticities, markups, and spillover effects are then calculated analytically according to the propositions derived in Section 3. I perform 100 bootstraps to estimate standard errors for the identified outcomes, treating firms as independent and resampling them with replacement.

6 Results

This section presents the empirical findings from the structural estimation. I begin by examining returns to scale (RTS) and the elasticity decomposition across factor inputs. I then analyze how markups vary with firm size and intangible intensity, and compare aggregate markup dynamics over time with and without intangible capital in the model. Finally, I discuss productivity levels and dispersion by firm size and intangible assets, and document the decline in knowledge diffusion over time.

6.1 Returns to Scale and Factor Input Elasticities

To examine how intangible capital affects firms' production, I estimate the production function under two specifications: one that includes intangible capital and knowledge spillovers, and one that excludes both variables. The specification with intangible capital is described in the Model section. The alternative specification is obtained by re-estimating the production function after removing the intangible capital input and the spillover term.

For each specification, I construct firm-level returns to scale (RTS) as the sum of the estimated output elasticities of all factor inputs. Because elasticities are recovered nonparametrically at the firm–year level using the control-variable approach, RTS are allowed to vary flexibly across firms, industries, and time. I then average firm-level RTS within industry–year size deciles to isolate systematic variation with scale while netting out industry-specific production technologies.

Figure 1 presents the RTS results. When intangible capital is included as an input in the production function, returns to scale increase monotonically with firm size, rising from approximately 1.1 in the smallest decile to about 1.6 in the largest decile. In contrast, when intangible capital is excluded, estimated RTS remain close to one and exhibit little variation across the size distribution.

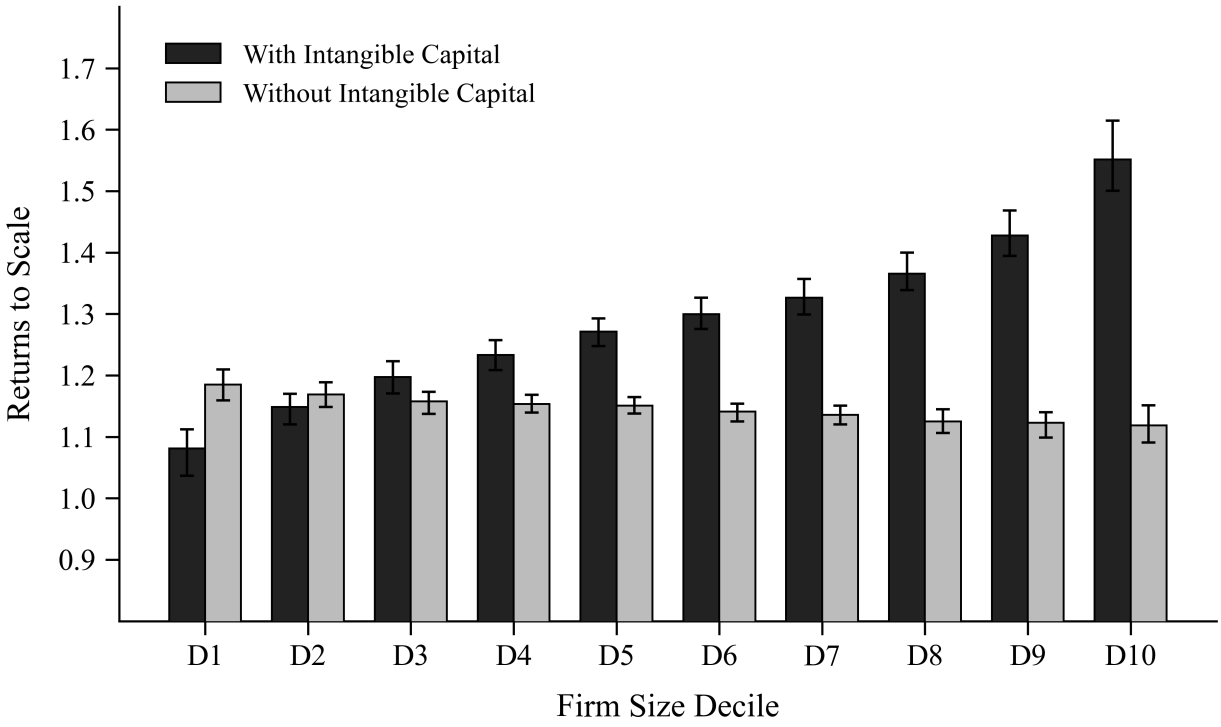
This comparison is central to the paper's mechanism. Increasing returns do not arise mechanically from physical scale; they emerge only when intangible capital enters the production technology. In the specification without intangibles, the technology is approximately constant-returns across firms, consistent with standard production-function estimates in the literature. Once intangible capital is incorporated, however, the curvature of the production function changes systematically with size.

Importantly, this pattern reflects technological scalability rather than changes in market power. RTS are constructed solely from output elasticities—identified from cost-minimization conditions and control variables—and are therefore independent of markup estimates. The evidence thus indicates that the observed rise in scale economies among large firms is a production-side phenomenon driven by asset composition, not by pricing distortions.

Figure 2 examines the time dimension of this relationship by estimating returns to scale separately for the pre-2000 and post-2000 periods. The figure makes clear that the size–RTS gradient is not stable over time. In both periods, returns to scale increase with firm size, consistent with the cross-sectional evidence in Figure 1. However, the magnitude and dispersion of scale economies change markedly after 2000. Prior to 2000, RTS increase gradually across size deciles, and the gap between the smallest and largest firms is modest. The production technology exhibits mild but broadly shared scale advantages.

After 2000, the gradient steepens substantially. Small firms experience a decline in returns

Figure 1: Returns to Scale by Firm Size Decile



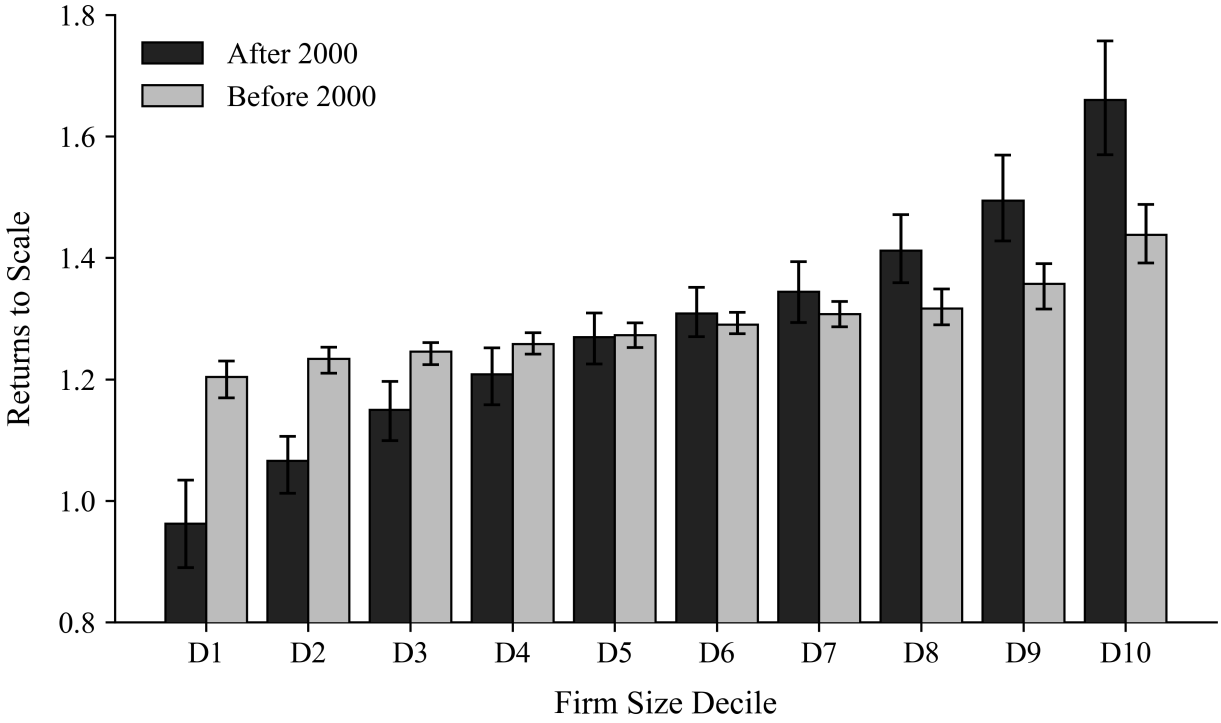
Note: This figure reports returns to scale by firm size decile. Estimates are averaged within industry-year cells and then across industry-years. Error bars denote 95% bootstrap confidence intervals (100 replications).

to scale, with the lowest decile moving closer to constant returns. In contrast, large firms exhibit a pronounced rise in scale economies, with the top decile reaching significantly higher RTS than in the earlier period. The widening gap between the bottom and top deciles indicates that dispersion in technological scalability increases over time.

Two features are worth emphasizing. First, the post-2000 pattern is not a uniform upward shift in scale economies. Instead, it is characterized by divergence: scale advantages strengthen among large firms while weakening among small firms. Second, because RTS are constructed purely from estimated output elasticities, this divergence reflects a change in the curvature of the production function rather than shifts in market power or demand conditions.

These results point to a structural shift in production technology over the last two decades. The growing importance of intangible capital—documented in the Data section and incorporated as a predetermined input in the model—appears to have altered the scalability of production in a highly asymmetric manner. The post-2000 environment amplifies the technological advantages of firms able to deploy intangible assets at scale, while offering

Figure 2: Returns to Scale by Firm Size Decile Before & After 2000



Note: This figure reports returns to scale by firm size decile. Estimates are averaged within industry-year cells and then across industry-years.

limited scale benefits to smaller firms. This asymmetric scalability provides a production-based mechanism for rising concentration and divergence across firms.

Figure 3 decomposes the size–RTS gradient into its underlying components by reporting firm-level output elasticities for each factor input across size deciles, separately before and after 2000. Because elasticities are identified from the nonparametric production function using the control-variable strategy, these patterns reflect technological responses rather than mechanical accounting identities.

Panel A (Intangible Capital). The elasticity of intangible capital increases sharply with firm size in both periods, and the slope becomes substantially steeper after 2000. For small firms, the estimated intangible elasticity is close to zero—and slightly negative in the post-2000 period—whereas for large firms it rises markedly, reaching economically large magnitudes in the top deciles. This pattern implies that the marginal product of intangible capital is highly size-dependent. In the post-2000 regime, intangible assets generate disproportionately higher output responses among large firms, consistent with scalable deployment of non-rival assets.

The negative or near-zero elasticity among the smallest firms should not be interpreted as implying that intangible investment destroys value per se. Rather, it indicates that,

conditional on other inputs and productivity, small firms are unable to translate intangible stocks into proportional output gains.[5] This is consistent with the presence of fixed costs, complementarities, or scale thresholds required to effectively exploit intangible assets.

Panel B (Physical Capital). The elasticity of physical capital increases modestly with firm size in both periods, but the magnitude is small relative to that of intangible capital. While there is a slight post-2000 increase among large firms and a mild decline among small firms, these changes are quantitatively limited. Physical capital therefore does not drive the steepening of the RTS gradient. This finding underscores that the source of increasing returns in the modern period is not traditional capital deepening but the accumulation of non-physical assets.

Panel C (Materials). The materials elasticity is largely flat across the size distribution and stable over time. Because materials are a flexible input identified from cost-minimization conditions, this stability suggests that intermediate input intensity does not explain the widening dispersion in scale economies. The production shift is therefore not attributable to changes in variable input usage.

Panel D (Labor). In contrast, labor elasticity declines systematically with firm size in both periods, and the downward slope steepens after 2000. Larger firms rely less on marginal labor input, conditional on effective labor productivity. Combined with the rise in intangible elasticity, this pattern indicates a reallocation of production importance away from labor and toward intangible capital as firms scale.

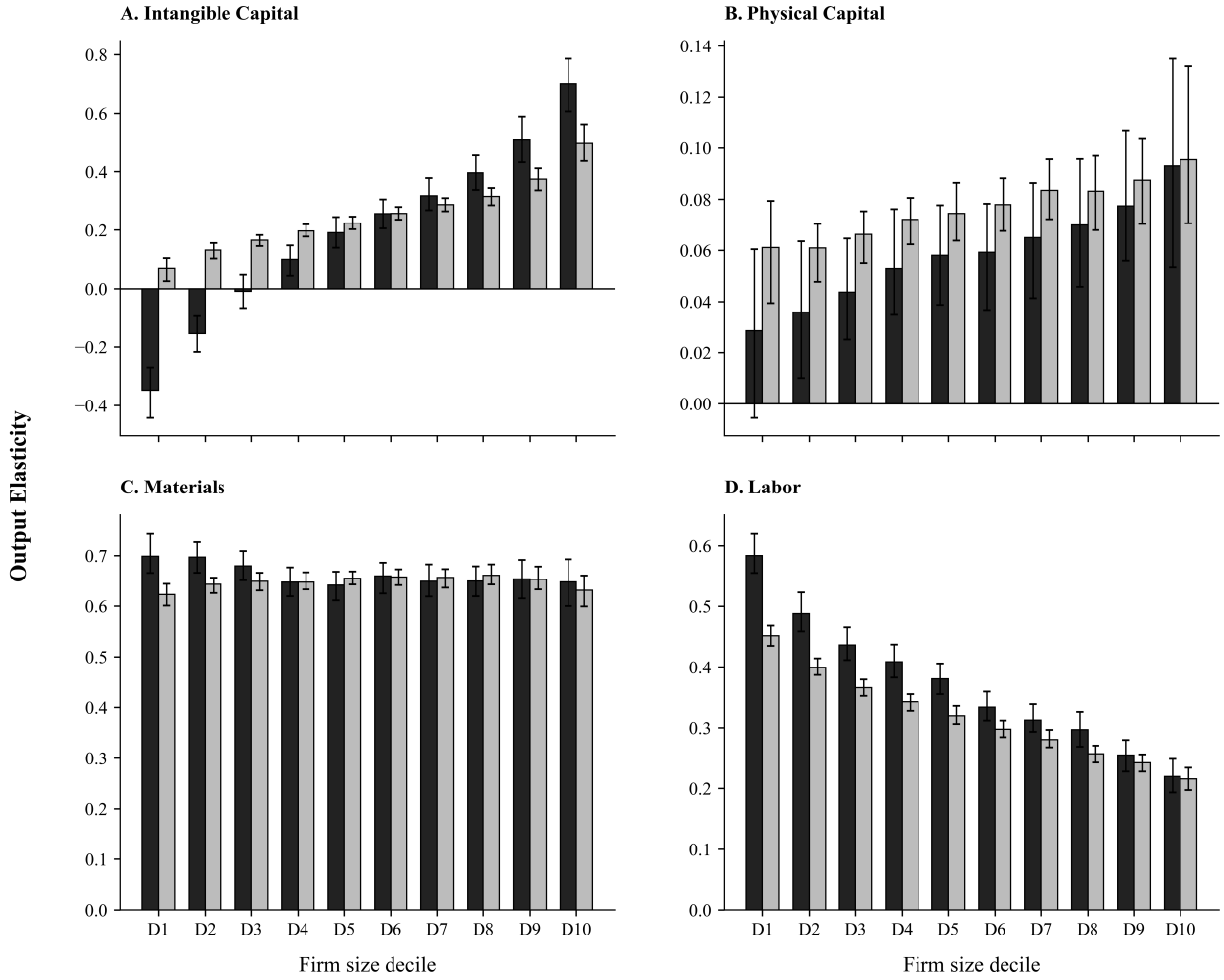
Figure 3, thus, clarifies the mechanism behind the post-2000 increase in returns to scale documented in Figures 1 and 2. The steepening size–RTS gradient is driven primarily by rising intangible elasticity among large firms, not by changes in tangible inputs. As firms grow larger, output becomes increasingly responsive to intangible capital and less responsive to labor. This reallocation in factor elasticities reflects a structural transformation in production technology: scalability in the modern era is rooted in the deployment of intangible assets rather than in physical scale or variable input intensity.

6.2 Markups

Having established that returns to scale vary systematically with firm size and over time, I now turn to markups. As described in the Identification section, firm-level markups are recovered from the first-order conditions of cost minimization following Loecker and Warzynski (2012). Specifically, the markup equals the ratio of a flexible input's output elasticity

[5] Elasticities are estimated conditional on the control variables that purge labor-augmenting and Hicks-neutral productivity. A negative elasticity therefore reflects low marginal contribution of intangible capital to contemporaneous output, not measurement error in revenue shares.

Figure 3: Factor Input Elasticities by Firm Size Decile



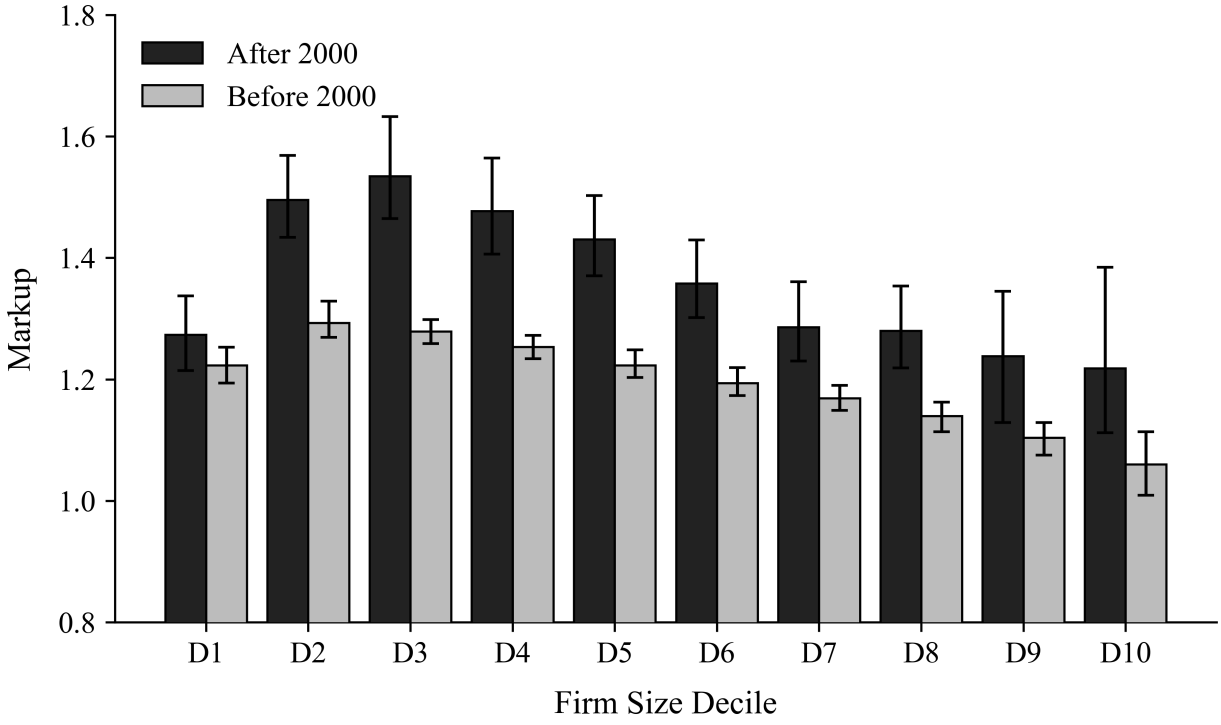
Note: This figure reports the output elasticities of intangible, physical capital, labor, and material by firm size decile. Estimates are averaged within industry-year cells and then across industry-years. Error bars denote 95% bootstrap confidence intervals (100 replications).
Note: Y-axis scales differ across panels.

to its revenue share. Because elasticities are identified non-parametrically and flexible-input shares are observed, markup variation reflects the interaction between estimated production elasticities and observed cost shares rather than imposed functional-form assumptions.[6]

Figure 4 reports average markups by firm size decile before and after 2000, averaging within industry–year cells and then across industry–years to abstract from industry-specific demand conditions. First, markups exhibit an inverted-U relationship with firm size in both periods. Markups rise from the smallest deciles to the middle of the size distribution and

[6]Since intangible capital is predetermined and does not enter the flexible-input first-order conditions, its elasticity affects markups only indirectly through the overall curvature of the production function, not mechanically through the markup formula.

Figure 4: Markup by Firm Size Decile Before & After 2000



Note: This figure reports markup by firm size decile. Estimates are averaged within industry-year cells and then across industry-years. Error bars denote 95% bootstrap confidence intervals (100 replications).

then decline for the largest firms. Pricing power is therefore not monotonically increasing in size. Medium-sized firms charge the highest margins, while both the smallest and the largest firms operate at lower markups. Second, markups shift upward after 2000 across nearly all size deciles. The level increase is particularly pronounced among small and medium-sized firms, though it is visible throughout the distribution. Importantly, the inverted-U shape remains intact. Thus, although scale economies become more concentrated among large firms after 2000, markups do not display a parallel monotonic increase at the top of the size distribution. Large firms experience stronger technological scalability but do not exhibit proportionally higher markups relative to mid-sized firms.

Taken together with the RTS results, this evidence highlights a key distinction between scalability and market power. Returns to scale capture the curvature of the production function, whereas markups capture the price–cost wedge. The post-2000 environment strengthens scale advantages at the top without generating a corresponding monotonic increase in pricing power. Technological scalability and market power therefore exhibit distinct size profiles.

Figure 5 sharpens this distinction by reporting markups across deciles of intangible intensity, again separately for the pre- and post-2000 periods. In contrast to the inverted-U

pattern by firm size, markups increase monotonically with intangible intensity. Firms in the lowest intangible decile have the lowest markups, while those in the highest decile exhibit substantially higher margins. This positive gradient is present in both periods and becomes noticeably steeper after 2000.

The post-2000 shift is especially pronounced among highly intangible-intensive firms. While markups rise broadly after 2000, the increase is largest at the top of the intangible distribution, leading to greater dispersion in pricing power across firms. Unlike the size-based distribution—where the largest firms do not command the highest markups—firms with greater intangible intensity consistently charge higher markups.

Considering this figure with Figures 4 and 5 indicate that markup behavior is more closely aligned with asset composition than with firm size per se. Although returns to scale increase strongly with size when intangible capital is included, pricing power follows intangible intensity much more tightly than scale. This reinforces the central mechanism of the paper: intangible capital reshapes both production technology and competitive outcomes, but along distinct margins. It generates increasing returns—especially among large firms—while simultaneously supporting higher markups among firms that are more intangible-intensive. Scale and market power are therefore related but not equivalent consequences of intangible accumulation.

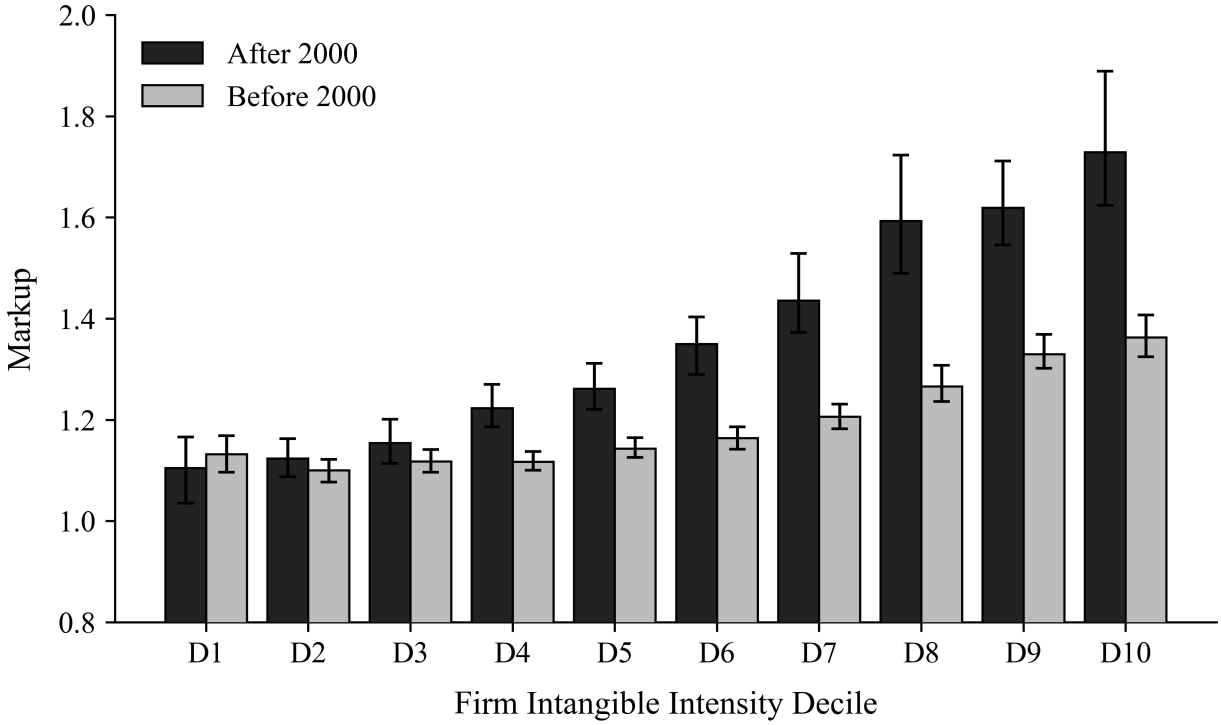
Figure 6 examines the time-series evolution of sales-weighted average markups under two alternative production specifications: one that includes intangible capital as a predetermined input and one that excludes it. In both specifications, average markups trend upward over the sample period, particularly after 2000. This pattern is consistent with the broader empirical literature documenting rising markups in recent decades (e.g., De Loecker et al. (2020)).

A key difference, however, emerges across specifications. Markups estimated from the model that excludes intangible capital are systematically higher than those obtained when intangible capital is included. The gap between the two series is persistent and, in some periods, widens over time.

This divergence has a structural interpretation. In the markup formula, markups are recovered as the ratio of flexible-input output elasticities to revenue shares. When intangible capital is omitted from the production function, part of the output response that is technologically attributable to intangible inputs is instead absorbed by the estimated elasticities of flexible inputs. Because these elasticities enter directly into the markup formula, the omission leads to an upward bias in measured markups. Put differently, failing to account for intangible capital misattributes technological scale effects to price–cost wedges.[7]

[7]This bias arises because, in the absence of intangible capital, the production function is misspecified.

Figure 5: Markup by Firm Intangible intensity Decile Before & After 2000



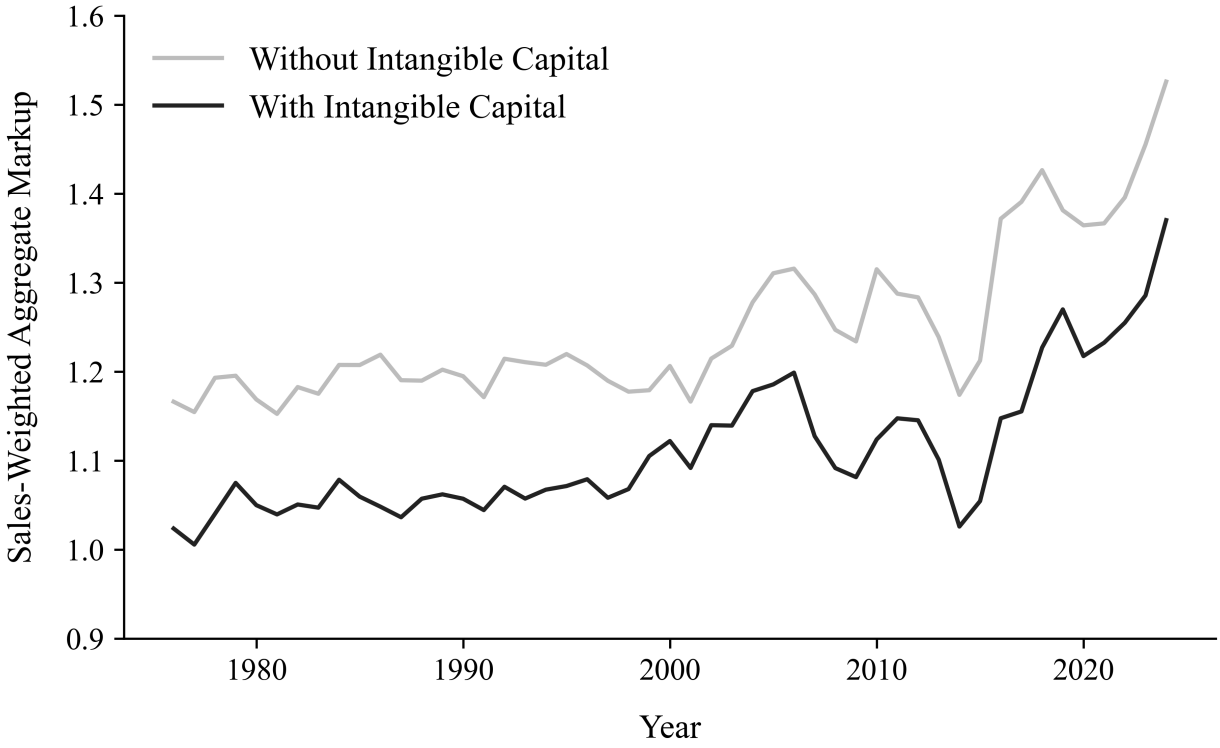
Note: This figure reports markup by firm size decile. Estimates are averaged within industry-year cells and then across industry-years. Error bars denote 95% bootstrap confidence intervals (100 replications).

The gap between the two series becomes more pronounced after 2000, precisely when intangible capital intensifies in the data and its estimated elasticity increases sharply among large firms. This timing strengthens the interpretation that part of the measured rise in markups reflects a shift in production technology rather than solely an expansion of market power.

Taken together with the cross-sectional evidence, the time-series results suggest that the post-2000 increase in aggregate markups is partly technological in origin. As intangible capital becomes a central input in production, models that omit it overstate market power by conflating scalability with monopoly pricing. Incorporating intangible capital therefore not only reshapes the estimated curvature of the production function but also refines the measurement of market power in the modern economy.

The control-variable procedure reallocates part of the output variation associated with intangibles to flexible inputs, inflating their estimated elasticities and, mechanically, the implied markup.

Figure 6: Average Markup Over Time With and Without Intangible Capital



Note: This figure reports markup over time. Estimates are averaged within industry-year cells and then across industry-years. Error bars denote 95% bootstrap confidence intervals (100 replications).

6.3 Productivity and Knowledge Diffusion

I next examine the diffusion margin embedded in the model. External knowledge spillovers enter the Hicks-neutral productivity process through the Markov structure in Section 2, and the spillover elasticity is identified as the derivative of the recovered productivity function with respect to the aggregate knowledge stock of other firms. Spillover exposure therefore measures how sensitive a firm's Hicks-neutral productivity is to external R&D, conditional on its own inputs and control variables. This approach connects directly to the innovation-spillover literature (e.g., Bloom et al. (2013); Akcigit and Ates (2023)) but embeds spillovers inside a structural production framework.

Figure 7 shows that before 2000, spillover exposure is relatively flat across the size distribution. Firms of different sizes exhibit broadly similar responsiveness of productivity to external knowledge stocks, with only modest variation across deciles. Knowledge diffusion in this period appears comparatively diffuse and consistent with earlier evidence of widespread spillovers across firms (Bernstein and Nadiri (1988); Bloom et al. (2013)).

After 2000, the pattern changes sharply. Spillover exposure declines monotonically with

firm size: small firms exhibit substantially higher external spillover elasticities, while large firms display markedly lower exposure. The divergence is particularly pronounced at the extremes of the size distribution. The smallest deciles experience a notable increase in spillover sensitivity, whereas the largest deciles exhibit a significant decline.

Because spillover elasticities are derived from the estimated productivity function—rather than inferred from citation networks or patent-based measures—this pattern reflects a structural change in how external knowledge affects firm-level production. The post-2000 environment is characterized by increasing reliance on external knowledge among small firms and decreasing reliance among large firms.

These results suggest a shift in the structure of knowledge diffusion after 2000. Whereas spillovers were broadly shared across firms in the earlier period, the post-2000 environment is characterized by greater reliance on external knowledge among small firms and reduced spillover exposure among large firms. This pattern is consistent with a production environment in which large firms increasingly internalize knowledge creation—potentially through their own intangible capital accumulation—while smaller firms rely more heavily on external sources of innovation. The reorganization of knowledge flows, therefore, complements the earlier findings on scale economies and markups, highlighting how the growing importance of intangible capital reshapes both production and diffusion dynamics across firms.

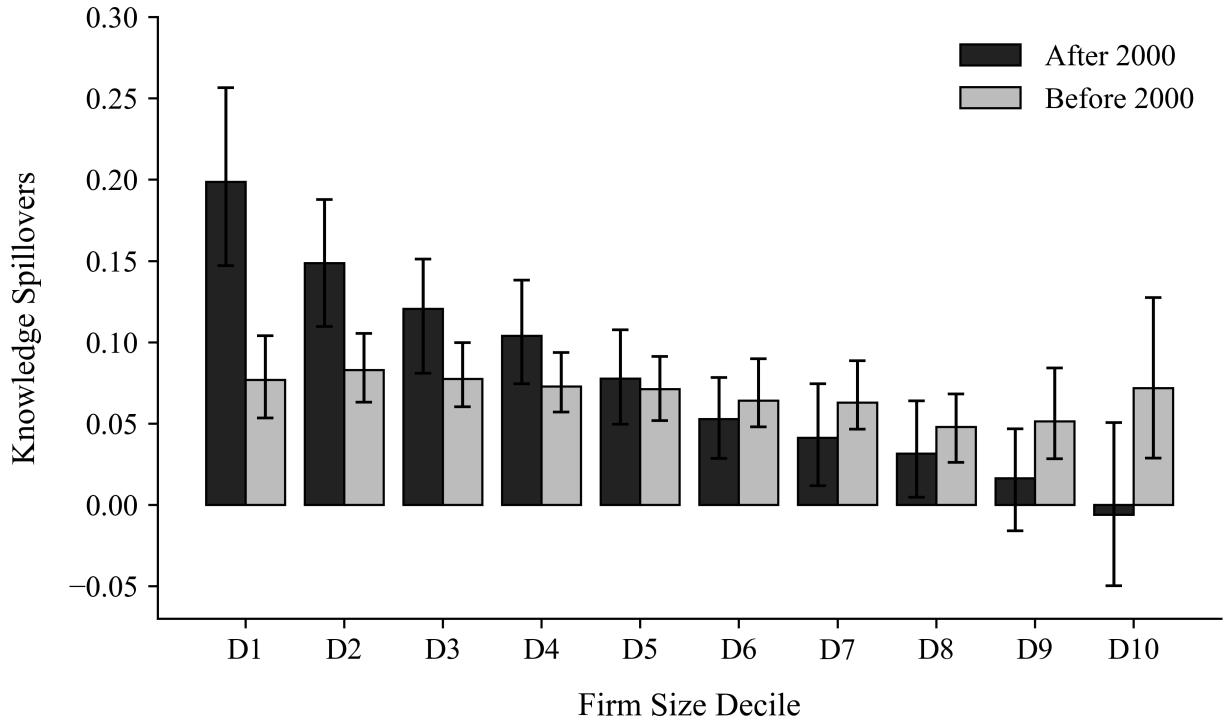
Figure 8 provides a complementary perspective by reporting industry–year demeaned productivity (2.5% trimmed) across joint deciles of firm size and intangible intensity.[8] Productivity corresponds to the recovered Hicks-neutral component net of industry–year effects, consistent with the identification strategy in Section 3. Deciles are constructed within industry–year cells using sales and intangible intensity, respectively. Each cell reports the average relative productivity of firms in that joint decile.

First, relative productivity declines systematically with firm size within industry–year cells. Moving from the smallest to the largest output deciles, demeaned productivity falls sharply. Large firms are therefore not more productive in the Hicks-neutral sense once industry and time effects are removed. This finding adds nuance to selection-based interpretations of firm growth (e.g., Syverson (2011); Hsieh and Klenow (2009)) and contrasts with models in which larger firms dominate purely because they are more efficient. This pattern indicates that size alone does not correspond to higher productivity once industry and time effects are removed.

Second, relative productivity increases monotonically with intangible intensity. Across all size categories, firms in higher intangible-intensity deciles exhibit higher demeaned productivity than those in lower deciles. Even among large firms, those that are more intangible-

[8] The untrimmed version yields nearly identical qualitative patterns and is reported in the Appendix.

Figure 7: Average Knowledge Spillover Before and After 2000 by Firm Size



Notes: The figure reports average external R&D spillover exposure by firm size decile before and after 2000. Estimates are first averaged within industry-year cells and then across industry-years. Error bars denote 95% bootstrap confidence intervals (100 replications).

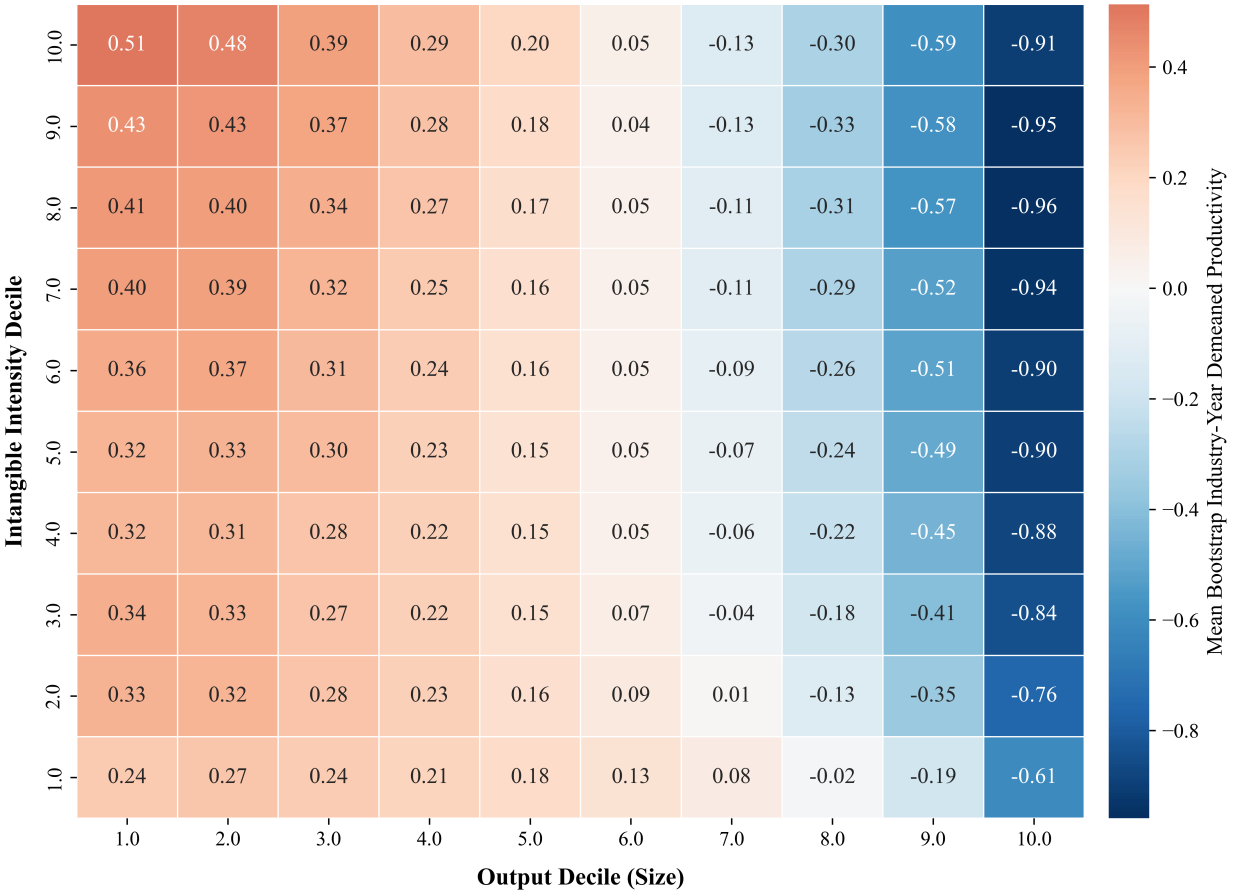
intensive display higher relative productivity than similarly sized firms with lower intangible intensity. This pattern is consistent with evidence that intangible capital enhances firm performance and value (Corrado et al. (2005); Eisfeldt and Papanikolaou (2014); Peters and Taylor (2017)).

The figure suggests that productivity advantages are more closely aligned with intangible intensity than with firm size per se. While larger firms exhibit stronger returns to scale, they are not necessarily more productive within industry–year once intangible intensity is accounted for. Instead, intangible intensity appears to be the key determinant of relative productivity.

This finding complements the earlier evidence: scale economies rise with size, markups rise with intangible intensity, and productivity advantages are concentrated among firms that accumulate intangible capital rather than simply those that are large.

This joint evidence integrates the paper's central findings and connects them to several literatures. Scale economies increase with firm size when intangible capital is included, consistent with the rise of "superstar" firms (Autor et al. (2020)); markups rise with in-

Figure 8: Industry-Year Relative Productivity by Firm Size and Intangible Intensity



Notes: The figure reports industry-year demeaned productivity (2.5% trimmed) by firm size and intangible intensity deciles. Deciles are formed within each industry-year using log output and intangible intensity, respectively. Each cell reports the unweighted mean productivity across firms in that decile.

tangible intensity rather than size; spillover exposure declines with size but increases with intangible intensity; and relative productivity is highest among intangible-intensive firms rather than among the largest firms per se. The rise of large firms in the modern economy therefore reflects a structural shift toward a production regime in which scalable intangible assets—rather than purely superior operational efficiency—govern firm growth, market structure, and knowledge diffusion.

7 Robustness Checks

7.1 Controlling for the Product Demand

One concern about the Compustat data is that the input and output variables are measured in revenue. In other words, the output in the regression becomes PQ where P is the price of product and Q is the quantity produced. Similarly, intangible capital is $K_{it}^{int} = P_t^{int} Q_{it}^{int}$, $K_{it} = P_t^k Q_{it}^k$ and $M_{it} = P_t^m Q_{it}^m$ where P_t^{int}, P_t^k, P_t^m are the price of intangible capital, capital and material, respectively. Note that as the input market is assumed to be perfectly competitive, the input price will be constant across the firms. By deflating the factor inputs, I control input prices, ensuring that the quantity of factor inputs is measured. However, since the output market is imperfectly competitive, even if I deflate the sales, the output price variation will not be captured. Thus, after deflating the variables, the estimation becomes

$$\frac{R_{it}}{P_t} = \frac{P_{it} Q_{it}}{P_t} = F_t(Q_{it}^k, Q_{it}^{int}, h_t(\omega_{it}^L L_{it}), Q_{it}^m) exp(\omega_{it}^H) exp(\epsilon_{it}) \tag{23}$$

where R_{it} the revenue of the firm i in year t.

I follow Klette and Griliches (1996) to control for the price variation among firms. Under the CES demand aggregator, the demand for each product is

$$Q_{it} = \left(\frac{P_{it}}{P_t}\right)^{-\sigma} Q_t exp(\epsilon_{it}^d) \tag{24}$$

where σ is the elasticity of substitution among the products, Q_t is the aggregate demand in the sector, and ϵ_{it}^d is an idiosyncratic demand shock. Rearranging the equation 24 the deflated output price becomes

$$\frac{P_{it}}{P_t} = \left(\frac{Q_{it}}{Q_t}\right)^{-1/\sigma} exp((\epsilon_{it}^d)^{1/\sigma}) \tag{25}$$

Substituting the equation 25 into the equation 23, the logarithm of the equation 23 becomes

$$r_{it} = (1 - \frac{1}{\sigma}) f_t(Q_{it}^k, Q_{it}^{int}, h_t(\omega_{it}^L L_{it}), Q_{it}^m) + (1 - \frac{1}{\sigma})\omega_{it}^H + \frac{1}{\sigma} q_t + \frac{1}{\sigma}\epsilon_{it}^d + (1 - \frac{1}{\sigma})\epsilon_{it} \tag{26}$$

where r_{it} is the logarithm of the deflated revenue, q_{it} is the logarithm of Q_{it} and q_t is the logarithm of Q_t. This equation states that when revenue is used for the production function estimation, the coefficient estimates become biased as we multiply the production function by $1 - \frac{1}{\sigma}$ and we have additional control variable which is the total industry demand. In order to isolate the interaction between demand and production coefficients, I estimate the

equation 26 as follows. I proceed similar estimation strategy as described in the estimation strategy section. I, additionally, include the size of sector q_t as an additional control variable. I first obtain the estimate for σ from this regression and then correct the elasticities by $1-\frac{1}{\sigma}$.

I re-estimate the structural model including q_t as an additional regressor to identify σ. This allows me to recover the "true" physical elasticities by scaling the revenue-based estimates by $(1-1/\sigma)^{-1}$. As shown in Table 1, the results remain remarkably stable. The RTS gradient and the markup divergence persist even after purging the demand-side contamination, suggesting that the "Technology of Scale" documented in this paper is a supply-side phenomenon rather than a byproduct of idiosyncratic pricing power or demand-side shocks.

Table 1: Robustness Check Results

	Returns to Scale		Knowledge Spillovers		Productivity Dispersion	
Specification	Δ_{50}	Δ_{10}	Δ_{50}	Δ_{10}	Δ_{50}	Δ_{10}
Baseline Model	0.205	0.455	−0.070	−0.127	−0.697	−1.594
Demand Control (CES)	0.201	0.449	−0.048	−0.086	−0.672	−1.537
Organizational Capital ($\gamma = 0.3$)	0.241	0.527	−0.043	−0.066	−0.659	−1.498
Measurement Error Simulation	0.038	0.086	−0.065	−0.116	−0.434	−0.898
Third-Order Polynomial	0.206	0.444	−0.066	−0.120	−0.670	−1.516
External Intangible Capital	0.199	0.450	−0.046	−0.085	−0.639	−1.475

Notes: Each row reports a separate robustness specification. Δ_{50} = top quintile minus bottom quintile (deciles 6–10 vs. 1–5); Δ_{10} = decile 10 minus decile 1. Returns to scale and knowledge spillovers are sales-share-weighted averages including intangible capital and inter-firm R&D spillovers. Productivity dispersion is the demeaned log productivity gap across quantiles.

7.2 Measurement Error in Intangible Capital

A fundamental empirical challenge in this framework is the potential mismeasurement of intangible capital. To ensure that the core findings are not driven by specific capitalization assumptions or idiosyncratic measurement noise, I conduct three distinct robustness exercises. First, I test the sensitivity of the model to the organizational capital parameters. Rather than applying the industry-varying capitalization rates estimated by Ewens et al. (2025), I impose a uniform capitalization rate of 30% ($\gamma = 0.3$) for SG&A expenditures across all sectors. This fixed 30% parameterization serves as a strict and widely utilized benchmark in the macroeconomic and finance literature to measure organizational capital.

Second, I explicitly simulate the impact of classical measurement error on the baseline intangible capital stock via a Monte Carlo approach. For each observation, I inject a normally

distributed noise term with a mean of zero and a standard deviation calibrated to 10% of the cross-sectional standard deviation of intangible capital within that specific industry-year cell. I generate 100 independent, perturbed datasets, re-estimate the structural production function for each sample, and compute the average of the resulting estimates.

Third, I account for the inorganic accumulation of intangible assets through mergers and acquisitions (M&A). When firms acquire targets, they inherently absorb the target's existing intangible capital stock. To prevent these discrete, externally acquired assets from confounding the firm's endogenous input choices, I incorporate acquired intangible capital as a structural shifter within the Markov productivity process. This explicitly controls for M&A-driven jumps in the state space, which are particularly prevalent among large incumbent firms.

All three robustness exercises yield results that are highly consistent with the baseline model. Specifically, returns to scale continue to strictly increase with firm size across all specifications. In the Monte Carlo simulation, the parameters are slightly attenuated relative to the main specification—a predictable econometric consequence of introducing classical measurement error. However, the main findings remain consistent with the baseline model. Furthermore, the core dynamics regarding firm-level markups and productivity spillover effects remain entirely robust to these perturbations. Detailed estimation results for all three exercises are reported in the Appendix.

7.3 Functional Form of the Approximations

To verify that the structural estimates are not artificially driven by the choice of polynomial degree, I test the sensitivity of the model to a more flexible functional form. While the baseline estimation approximates the unknown production and productivity functions using second-order polynomials, I re-estimate the unknown production function employing third-order polynomial expansions.

The core findings prove highly robust to this increased nonparametric flexibility. Specifically, the estimated output elasticities, firm-level markups, and knowledge spillover effects remain both quantitatively and qualitatively consistent with the baseline results. The detailed estimates for this higher-order specification are reported in the Appendix.

7.4 External Validation and Structural Falsification

To formally address the potential for measurement bias, I validate the structural markup estimates against Tobin's Q, a market-based measure of firm value that is entirely external to the production function estimation. The central identification concern is a measurement

tautology: including intangible capital as an additional factor input may mechanically reduce the estimated output elasticities of flexible inputs, and because markups are recovered as the ratio of these elasticities to revenue shares, any such reduction would lower measured markups by construction. If this mechanical adjustment were the sole driver of the corrected estimates, one would expect the model to exhibit a weaker or unchanged correlation with external market valuations — a lower markup that reflects nothing more than an accounting reallocation should not command a stronger relationship with forward-looking investor assessments of firm value.

Table **??** reveals a pattern that is difficult to reconcile with a pure measurement artifact. Before 2000, when production was predominantly physical-capital-intensive and intangible stocks varied modestly across the size distribution, the two specifications yield nearly identical results. The corrected model provides a marginal improvement of 1.60% in R^2, and the markup coefficients in columns (1) and (2) are statistically indistinguishable. This near-equivalence serves as a falsification benchmark: in an environment where omitting intangibles generates little bias, the correction produces little change.

A sharp divergence emerges after 2000. In the modern intangible-intensive regime, the specification incorporating intangible capital yields an 11.51% improvement in $2R^2$ R2 relative to the traditional baseline, with the markup coefficient rising from 0.614 to 0.654. Despite being lower in absolute magnitude, corrected markups explain substantially more of the cross-sectional variation in market valuations. This is the opposite of what a mechanical artifact would produce: a purely accounting-driven reduction in elasticities should weaken, not strengthen, the connection between estimated markups and external valuations. The result instead suggests that traditional estimators conflate technological scale effects with pricing power, and that once intangible capital is properly accounted for, the residual markup variation more accurately reflects the genuine price-cost wedge that investors price into firm value.

8 Conclusion

Over the past decades, the structural foundation of the modern economy has fundamentally shifted from physical-capital-intensive production toward the accumulation of intangible assets. This paper develops a flexible, nonparametric production function framework to evaluate how this transition has reshaped firm-level scale economies, market power, and knowledge diffusion. By explicitly incorporating intangible capital as a distinct, non-rival factor input and external R&D as a structural productivity shifter, the model overcomes the mechanical limitations of rigid parametric specifications. This approach provides a robust

mechanism to jointly recover heterogeneous factor elasticities, markups, and spillover effects while accommodating both labor-augmenting and Hicks-neutral unobserved productivity.

The central finding of this paper is that the increasing returns to scale characteristic of modern "superstar" firms are not a mechanical byproduct of sheer physical size, but are instead fundamentally driven by intangible asset accumulation. Once intangible capital is properly specified within the production technology, returns to scale rise monotonically with both firm size and intangible intensity. Furthermore, I demonstrate that while intangible-intensive firms are highly scalable, they are not necessarily more productive in a traditional Hicks-neutral sense. In fact, within industry-year cells, Hicks-neutral productivity strictly declines with firm size. This reveals a critical nuance in modern industrial dynamics: dominant firms leverage the non-rivalry of their intangible assets to expand output disproportionately relative to standard inputs, driving industry concentration through hyper-scalability rather than pure technological superiority.

This intangible-driven scalability fundamentally alters firm pricing behavior and the diffusion of aggregate knowledge. I find that firm-level markups exhibit an inverted-U relationship with firm size, suggesting that the largest firms exploit their immense scale economies to pursue a volume-margin trade-off, capturing market share through lower markups rather than extracting maximum price-cost margins. Moreover, while standard estimators routinely overstate market power by misattributing these technological scale effects to pricing power, explicitly modeling intangible capital significantly lowers aggregate markup estimates. Concurrently, I document that as firms grow larger and accumulate massive internal intangible stocks, their reliance on external knowledge spillovers systematically declines.

Ultimately, these findings indicate that the rise of dominant firms, the secular decline in the labor share, and the weakening of aggregate knowledge diffusion are all interconnected symptoms of a deeper technological transformation. In an era where production is increasingly governed by codified knowledge, proprietary data, and organizational capital, traditional models that omit intangible assets critically mischaracterize the drivers of firm growth. This framework offers a highly tractable, empirically rigorous foundation for future research exploring the evolving intersections of innovation, market structure, and macroeconomic inequality.

Appendix

A Proofs

A.1 Proof of Equation 3

The firms minimize their flexible inputs as physical and intangible capital are predetermined variables. They optimize their level to produce at least planned output, $\bar{Y}_{it}$. The cost minimization problem, then, becomes:

$$\min_{L_{it},M_{it}} \left\{p_t^L L_{it} + p_t^M M_{it} : \mathbb{E}[F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it}))exp(\omega_{it}^H)exp(\epsilon_{it})|\mathcal{I}_{it}] \geq \bar{Y}_{it}\right\} \tag{27}$$

By the timing assumption, as capitals and ω_{it} and ω_{it}^L are known, the cost problem turns out

$$\min_{L_{it},M_{it}} \left\{p_t^L L_{it} + p_t^M M_{it} : F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it}))exp(\omega_{it}^H)\mathbb{E}[exp(\epsilon_{it})\mathcal{I}_{it}] \geq \bar{Y}_{it}\right\}$$

This problem can be written as the choice of effective labor rather than labor. $\bar{L}_{it} = L_{it}\omega_{it}^L$ is defined as the effective labor. Defining quality adjusted wages $p_{it}^L = \frac{p_t^L}{\omega_{it}^L}$ I can now rewrite the problem in terms of effective labor, $\bar{L}_{it}$

$$\min_{\bar{L}_{it},M_{it}} \left\{\bar{p}_{it}^L \bar{L}_{it} + p_t^M M_{it} : F_t(K_{it}, K_{it}^{int}, h_t(\bar{L}_{it}, M_{it}))exp(\omega_{it}^H)\mathbb{E}[exp(\epsilon_{it})] \geq \bar{Y}_{it}\right\} \tag{28}$$

These two problems are equivalent as I just redefine the optimization problem with effective labor and labor productivity ω_{it}^L is known by the firms at time t. For easy notation, defining $\tilde{Y}_{it} = \frac{\bar{Y}_{it}}{\mathbb{E}[exp(\epsilon_{it}\mathcal{I}_{it}]exp(\omega_{it}^H)}$ the cost problem becomes

$$C_t(K_{it}, K_{it}^{int}, \bar{Y}_{it}, p_{it}^L, p_t^M, \omega_{it}^H) = \min_{\bar{L}_{it},M_{it}} \left\{\bar{p}_{it}^L \bar{L}_{it} + p_t^M M_{it} : F_t(K_{it}, K_{it}^{int}, h_t(\bar{L}_{it}, M_{it})) \geq \tilde{Y}_{it}\right\}$$

As $F_t()$ is a homogeneous function in $h_t()$, I can invert $F_t()$ function in $h_t()$. As the objective function is linear I can reconstruct the problem as

$$C_t(K_{it}, K_{it}^{int}, \bar{Y}_{it}, p_{it}^L, p_t^M, \omega_{it}^H) = \min_{\bar{L}_{it},M_{it}} \left\{\bar{p}_{it}^L \bar{L}_{it} + p_t^M M_{it} : h_t(\bar{L}_{it}, M_{it}) \geq F_t^{-1}(K_{it}, K_{it}^{int}, \tilde{Y}_{it})\right\}$$

As $K_{it}, K_{it}^{int}, \omega_{it}^H$ and $\bar{Y}_{it}$ is given at time t, the problem can be rewritten as

$$C_t(K_{it}, K_{it}^{int}, \bar{Y}_{it}, p_{it}^L, p_t^M, \omega_{it}^H) = F_t^{-1}(K_{it}, K_{it}^{int}, \tilde{Y}_{it}) \min_{\bar{L}_{it},M_{it}} \left\{\bar{p}_{it}^L \bar{L}_{it} + p_t^M M_{it} : h_t(\bar{L}_{it}, M_{it}) \geq 1\right\}$$

Therefore, the cost minimization problem pins down to

$$C_t(K_{it}, K_{it}^{int}, \bar{Y}_{it}, p_{it}^L, p_t^M, \omega_{it}^H) = C_{1t}(K_{it}, K_{it}^{int}, \bar{Y}_{it}, \omega_{it}^H) C_{2t}(p_{it}^L, p_t^M)$$

where I redefine $F_t^{-1}()$, $C_{1t}(K_{it}, K_{it}^{int}, \bar{Y}_{it}, \omega_{it}^H) = F_t^{-1}(K_{it}, K_{it}^{int}, \tilde{Y}_{it}) \min_{\bar{L}_{it}, M_{it}}$. C_{2t} is the results of minimization problem. Since p_{it}^L and p_t^M are the only state variables left under the cost minimization problem, the minimization problem will only be a function of them. In other words, minimization part will be $C_{2t}(p_{it}^L, p_t^M)$.

Using Shephard's Lemma, the firms' optimal material and labor demand will be a derivative of the minimized cost function with respect to the corresponding prices, i.e.

$$M_{it} = \frac{\partial C_t(K_{it}, K_{it}^{int}, \bar{Y}_{it}, p_{it}^L, p_t^M, \omega_{it}^H)}{\partial p_t^M} = C_{1t}(K_{it}, K_{it}^{int}, \bar{Y}_{it}, \omega_{it}^H) \frac{\partial C_{2t}(p_{it}^L, p_t^M)}{\partial p_t^M}$$
$$\bar{L}_{it} = \frac{\partial C_t(K_{it}, K_{it}^{int}, \bar{Y}_{it}, p_{it}^L, p_t^M, \omega_{it}^H)}{\partial p_{it}^L} = C_{1t}(K_{it}, K_{it}^{int}, \bar{Y}_{it}, \omega_{it}^H) \frac{\partial C_{2t}(p_{it}^L, p_t^M)}{\partial p_{it}^L}$$

As the first part of the cost minimization problem, $C_{1t}(K_{it}, K_{it}^{int}, \bar{Y}_{it}, \omega_{it}^H)$ does not depend on flexible input prices, I factor out when taking derivatives. I will only need to take the derivative of $C_{2t}(p_{it}^L, p_t^M)$ to find the flexible input demand. Taking the ratio between material and labor demand, the equations above becomes

$$\frac{M_{it}}{\bar{L}_{it}} = \frac{\partial C_{2t}(p_{it}^L, p_t^M)/\partial p_t^M}{\partial C_{2t}(p_{it}^L, p_t^M)/\partial p_{it}^L}$$

The material-to-effective labor ratio only depends on the factor input prices. Replacing the effective labor $\bar{L}_{it} = L_{it}\omega_{it}^L$ into the equation, material-to-labor ratio becomes

$$\frac{M_{it}}{L_{it}} = \frac{\partial C_{2t}(p_{it}^L, p_t^M)/\partial p_t^M}{\partial C_{2t}(p_{it}^L, p_t^M)/\partial p_{it}^L} \omega_{it}^L$$

Note that this function only depends on the input prices and labor productivity, ω_{it}^L. Thus, the flexible input ratio becomes

$$\frac{M_{it}}{L_{it}} = r_t(p_t^L, p_t^M, \omega_{it}^L) = r_t(\omega_{it}^L)$$

where $r_t()$ is an unknown function. As $r_t()$ function is time-varying, and input prices are constant across firms due to the competitive input market, these prices will be a constant in this function. Therefore, the material-to-labor ratio becomes a function of labor augmenting productivity ω_{it}^L.

A.2 Proof of Equation 4

I will now show that the material-to-labor ratio is monotonous in ω_{it}^L in order to write labor productivity as a function of the material-to-labor ratio.

Since the cost function is a homogeneous of degree with one, the derivative of the cost function with respect to the input prices would be homogeneous of degree zero. Using this property, dividing by p_t^M will not change the partial derivative of the cost function with respect to flexible inputs. Therefore, I can rewrite it as follows

$$\frac{M_{it}}{L_{it}} = \frac{\partial C_{2t}(\frac{p_{it}^L}{p_t^M}, \frac{p_t^M}{p_t^M})/\partial p_t^M}{\partial C_{2t}(\frac{p_{it}^L}{p_t^M}, \frac{p_t^M}{p_t^M})/\partial p_{it}^L}\omega_{it}^L = \frac{\partial \tilde{C}_{2t}(\frac{p_{it}^L}{p_t^M})/\partial p_t^M}{\partial \tilde{C}_{2t}(\frac{p_{it}^L}{p_t^M})/\partial p_{it}^L}\omega_{it}^L$$

where $\tilde{C}_{2t}(\frac{p_{it}^L}{p_t^M}) = C_{2t}(\frac{p_{it}^L}{p_t^M}, 1)$. For the sake of easy notation, let $\tilde{C}_{2t}(\frac{p_{it}^L}{p_t^M})/\partial p_t^M = \tilde{C}_{2m}$, $\tilde{C}_{2t}(\frac{p_{it}^L}{p_{it}^L})/\partial p_t^M = \tilde{C}_{2l}$, and $\tilde{M}_{it} = \frac{M_{it}}{L_{it}}$. Taking the logarithm of the equation I can rewrite it as

$$log(\tilde{M}_{it}) = log\left(\frac{\tilde{C}_{2m}}{\tilde{C}_{2l}}\right) + log(\omega_{it}^L)$$

Taking the derivative of the equation with respect to $log(\omega_{it}^L)$

$$\frac{\partial log(\tilde{M}_{it})}{\partial log(\omega_{it}^L)} = \frac{\partial \tilde{C}_{2m}/\tilde{C}_{2l}}{\partial log(p_{it}^L/p_t^M)}\frac{log(\partial p_{it}^L/p_t^M)}{\partial log(\omega_{it}^L)} + 1$$

Since $\frac{log(\partial p_{it}^L/p_t^M)}{\partial log(\omega_{it}^L)} = 1$ and $\frac{\partial \tilde{C}_{2m}/\tilde{C}_{2l}}{\partial log(p_{it}^L/p_t^M)} = -\sigma(\omega_{it}^L L_{it}, M_{it})$ is the elasticity of substitution between effective labor and material, the equation becomes

$$\frac{\partial log(\tilde{M}_{it})}{\partial log(\omega_{it}^L)} = -\sigma(\omega_{it}^L L_{it}, M_{it}) + 1$$

From the elasticity substitution assumption $\sigma(\omega_{it}^L L_{it}, M_{it}) > 1$ or $\sigma(\omega_{it}^L L_{it}, M_{it}) < 1$, the derivative of $\tilde{M}$ with respect to ω_{it}^L is either positive or negative. IT will never become zero. This implies that $\tilde{M}_{it}$ is strictly increasing or decreasing function of ω_{it}^L. Thus, $\tilde{M}_{it}$ is monotone in ω_{it}^L. Then, we can invert the $r_t(\omega_{it}^L)$ function in ω_{it}^L, then labor productivity, ω_{it}^L becomes

$$\omega_{it}^L = r_t^{-1}\left(\frac{M_{it}}{L_{it}}\right) = r_t^{-1}(\tilde{M}_{it})$$

Thus, this shows that labor productivity is a function of the material-to-labor ratio.

A.3 Proof of Equation 11

The cost minimization problem of the firm i at time t is

$$\min_{L_{it},M_{it}} \left\{p_t^L L_{it} + p_t^M M_{it} : F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it}))exp(\omega_{it}^H)\mathbb{E}[exp(\epsilon_{it})\mathcal{I}_{it}] \geq \bar{Y}_{it}\right\}$$

The first order condition for material and labor is

$$p_t^M = \lambda_{it}\frac{\partial F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it}))exp(\omega_{it}^H)\mathbb{E}[exp(\epsilon_{it})\mathcal{I}_{it}]}{\partial M_{it}}$$
$$p_t^L = \lambda_{it}\frac{\partial F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it}))exp(\omega_{it}^H)\mathbb{E}[exp(\epsilon_{it})\mathcal{I}_{it}]}{\partial L_{it}}$$

where λ_{it} is the Lagrange multiplier on the constraint. λ_{it} will capture the firms' marginal cost. Multiplying the first equation by $\frac{M_{it}}{p_{it}Y_{it}}$ and the second equation by $\frac{M_{it}}{p_{it}Y_{it}}$

$$\frac{p_t^M M_{it}}{p_{it}Y_{it}} = \lambda_{it}\frac{\partial F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it}))exp(\omega_{it}^H)\mathbb{E}[exp(\epsilon_{it})\mathcal{I}_{it}]}{\partial M_{it}}\frac{M_{it}}{p_{it}Y_{it}}$$
$$\frac{p_t^L L_{it}}{p_{it}Y_{it}} = \lambda_{it}\frac{\partial F_t(K_{it}, K_{it}^{int}, h_t(\omega_{it}^L L_{it}, M_{it}))exp(\omega_{it}^H)\mathbb{E}[exp(\epsilon_{it})\mathcal{I}_{it}]}{\partial L_{it}}\frac{L_{it}}{p_{it}Y_{it}}$$

The right-hand side of the first equation becomes the material share in revenue. Similarly, the right-hand side of the second equation becomes labor share in revenue. Let's denote them as α_{it}^M and α_{it}^L as the material and labor share in revenue, respectively. On the left-hand side of the equations, $\frac{\partial F_t()}{\partial M_{it}}\frac{M_{it}}{Y_{it}}$ is the output elasticity of material, and similarly $\frac{\partial F_t()}{\partial L_{it}}\frac{L_{it}}{Y_{it}}$ is the output elasticity of labor. Let's denote ϵ_{it}^L, and ϵ_{it}^M as the labor and material elasticity, respectively. Thus, we have

$$\alpha_{it}^M = \frac{\lambda_{it}}{p_{it}}\epsilon_{it}^M$$
$$\alpha_{it}^L = \frac{\lambda_{it}}{p_{it}}\epsilon_{it}^L \tag{29}$$

Taking their ratios we will have

$$\frac{\alpha_{it}^M}{\alpha_{it}^L} = \frac{\epsilon_{it}^M}{\epsilon_{it}^L} \tag{30}$$

This shows that the ratio of revenue shares of flexible inputs is equal to their ratio of output elasticities.

A.4 Proof of Equation 13

In the equation 31, $\frac{\lambda_{it}}{p_{it}}$ is equal to the inverse markup because the markup is defined as price over marginal cost. Since λ_{it} is the marginal cost of the firm markup becomes $\mu_{it} = \frac{p_{it}}{\lambda_{it}}$ by definition. Therefore, we can rewrite equation 31 as

$$\begin{aligned} \alpha_{it}^{M} &= \frac{\epsilon_{it}^{M}}{\mu_{it}} \\ \alpha_{it}^{L} &= \frac{\epsilon_{it}^{L}}{\mu_{it}} \end{aligned} \tag{31}$$

Rearranging the equations markups can be rewritten as

$$\mu_{it} = \frac{\epsilon_{it}^{M}}{\alpha_{it}^{M}} = \frac{\epsilon_{it}^{L}}{\alpha_{it}^{L}}$$

The markup charged by the firms equals the ratio of their flexible input elasticity to the flexible input's share in revenue.

A.5 Proof of Proposition 1

For the production function,

$$y_{it} = f_t(K_{it}, K_{it}^{int}, L_{it} h_t(\omega_{it}^{L}, \tilde{M}_{it})) + \omega_{it}^{H} + \epsilon_{it}$$

The material and labor elasticity are

$$\begin{aligned} \theta_{it}^{M} &= f_{t3} h_{t2} M_{it} \\ \theta_{it}^{L} &= f_{t3}(h_t - h_{t2}\tilde{M}_{it}) L_{it} \end{aligned}$$

where f_{t3} is the derivative of f_t function with respect to the third arguments and h_{t2} is the derivative of h_t function with respect to the second argument. The sum of labor and material elasticity is

$$\theta_{it}^{M} + \theta_{it}^{L} = f_{t3} h_{t2} M_{it} + f_{t3}(h_t - h_{t2}\tilde{M}_{it}) L_{it} = f_{t3} h_t L_{it} = f_{t3} \bar{h}_t L_{it}$$

Since the sum of elasticities do not depend on the derivative of h_t function I can replace $h_t()$ with $\bar{h}_t()$. The last equality comes from that fact.

B Data and Estimation

B.1 Estimation Strategy

This section provides a detailed estimation strategy employed. I first construct the control variables, u^1_{it} and u^2_{it} to control for Hicks-neutral and labor augmenting productivity. u^1_{it} is defined in equation 7 as

$$u^1_{it} = F_{\tilde{M}_{it}|W_{it-1},\sum_{j\in I,j\neq i} R_{jt-1}}\Big(\tilde{M}_{it} \mid W_{it-1}, \sum_{j\in I,j\neq i} R_{jt-1}\Big)$$

u^1_{it} is computed as a conditional CDF of $\tilde{M}$ given $W_{it-1}, \sum_{j\in I,j\neq i} R_{jt-1}$. However, the dependent variable, $\tilde{M}_{it}$ is a continuous variable so I can't directly apply the logistic regression. To do that, I first split the data into year and sector subsets since I do production function estimation at the sector year level. I then partition the sub-sample data (within a year and sector) into 500 parts. In cases where 500 partitioning is not feasible I partition them into the one tenth of the sample size[9]. At the boundary points of each discretized data points (named as $q \in Q$) I construct the logistic regression as follow

$$P(\tilde{M}_{it} \leq q \mid W_{it-1} = w, \sum_{j\in I,j\neq i} R_{jt-1} = r) = s_{1t}(q, w, r)$$

The dependent variable becomes a discrete variable depending on whether $\tilde{M}_{it}$ is less than q. I approximate the s_{2t} function with the second-order polynomials[10]. I, then, estimate the predicted CDF for each q points using logistic regression. I compute the remaining points using a linear interpolation.

To estimate the u^2_{it} I follow the similar strategy as u^1_{it}. The equation 9 defines u^2_{it} as follow

$$u^2_{it} = F_{M_{it}|K_{it},K^{int}_{it},W_{it-1},\sum_{j\in I,j\neq i} R_{jt-1}}\Big(M_{it} \mid K_{it}, K^{int}_{it}, W_{it-1}, \sum_{j\in I,j\neq i} R_{jt-1}\Big)$$

As in u^1_{it}, the dependent variable M_{it} is not a discrete variable, it is a continuous variable. I partition the data 500 parts at the sector year level. If the data is not feasible I split it into the one-tenth of the data. At each q level, I do

$$P(M_{it} \leq q \mid K_{it} = k, K^{int}_{it} = k^{int}, W_{it-1} = w, \sum_{j\in I,j\neq i} R_{jt-1} = r, u^1_{it} = u^1) = s_{2t}(q, k, k^{int}, w, r, u^1)$$

[9] I use different partition size such as 5. They don't change the results

[10] Third-order approximation gives the same results.

The dependent variable becomes a discrete variable depending on whether M_{it} is less than q. I similarly approximate s_{2t} function with second order polynomials. I then find the CDF at the points where $M_{it} = q$. For the remaining points, I compute the linear interpolation so that I have an estimator for u^2_{it} at each point in the data.

After estimating u^1_{it} and u^2_{it}, I can now estimate the productions functions. I estimate the productions at each sector over 7-year rolling windows in the equation 19

$$y_{it} = f_t(K_{it}, K^{int}_{it}, L_{it}\bar{h}_t(\tilde{M}_{it})) + c_{2t}\big(W_{it-1}, \sum_{j\in I, j\neq i} R_{jt-1}, u^1_{it}, u^2_{it}\big) + \epsilon_{it}$$

If I know $\bar{h}_t(.)$ function, I can estimate the regression using polynomial approximations for $f_t()$ and $c_{2t}()$ functions. I first approximate the $h_t()$ function with third-order polynomials.

$$h_t(\tilde{m}_{it}) = \alpha_{1t}\tilde{m}^2_{it} + \alpha_{3t}\tilde{m}^3_{it}$$

I approximate the $f_t()$ function with second-order polynomials. For the sake of easy notation, let's define $v_{it} = l_{it} + h(m_{it})$. With the approximated $h_t()$ given, I can write $f_t()$ as follow

$$\begin{aligned} f_t() = \beta_{0t} + \beta_{1t}k_{it} \quad &+ \quad \beta_{2t}k^{int}_{it} + \beta_{3t}v_{it} + \beta_{4t}k^2_{it} + \beta_{5t}k_{it}k^{int}_{it} + \beta_{6t}k_{it}v_{it} \\ &+ \quad \beta_{7t}(k^{int}_{it})^2 + \beta_{8t}k^{int}_{it}v_{it} + \beta_{9t}v^2_{it} \end{aligned}$$

where β's represent the coefficients of the polynomial approximations. I have time subscripts because I estimate them over 8 year rolling windows within each sector. These coefficients will be sector-specific and time varying. I similarly approximate c_{2t} and c_{3t} with second-order polynomials. I do then minimize the moment conditions with these approximated functions in equation 22.

I estimate the production function parameters β's and $c_2()$ and $c_3()$ coefficients given α's. I first guess α values and then estimate the moment conditions using the least squares. I iterate the loops to minimize the moment conditions in equation 22. This will give me the β coefficients and productivity function coefficients. This is not a very hard loop because given *alphas* it is straightforward to find other parameters. Therefore, in the loops we are minimizing the OLS estimates to find 3 α parameters to minimize the residuals.

After estimating the parameters of $f_2()$ and c_2 functions, I can compute the elasticities as defined in the identification section. The output elasticity of intangible and tangible capital is the derivative of estimated f_t function with respect to the capital and intangible capital, respectively. Since I know that the approximated the f_t function the capital and intangible

elasticity are

$$\theta_{it}^{k} = \beta_{1t} + 2\beta_{4t}k_{it} + \beta_{5t}k_{it}^{int} + \beta_{6t}v_{it}$$
$$\theta_{it}^{int} = \beta_{2t} + \beta_{5t}k_{it}^{int} + 2\beta_{7t}k_{it}^{int} + \beta_{8t}v_{it}$$

Due to the input heterogeneity this generates elasticity heterogeneity across firms. For the identification of flexible inputs I similarly take the derivative of f_t function with respect to v, this would give me the sum of flexible input elasticities. Multiplying with the flexible share in revenue I compute the labor and material elasticities. The productivity elasticity of the external knowledge is the derivative of c_2 function with respect to external knowledge capital. I construct them similar to the output elasticities.

B.2 Data

	Observation	Mean	Std. Dev.	Min	p25	Median	p75	Max
sale	139475	2270.9	12878.0	0.3	40.7	189.8	923.0	680985.0
intangible	139475	571.1	3445.4	0.5	13.7	50.5	210.8	265123.8
capital	139475	1569.7	10242.1	0.5	17.3	93.2	501.1	477185.0
emp	139475	8.9	40.7	0.0	0.3	1.2	5.1	2300.0
metarial	139475	1790.5	11769.4	0.3	20.7	118.4	618.3	526207.6
fyear	139475	1998.5	12.7	1976.0	1988.0	1998.0	2008.0	2024.0

Notes: This table reports summary statistics for the primary variables used in the construction of the production function. All variables are deflated corresponsing deflators at BEA.

I use the Compustat data from Standard and Poor's Compustat North America database from 1975 to 2020. The Compustat data covers all public firms in the US. The data is available before 1975, but since intangible asset's parameters are estimated using a data from 1975 in Ewens et al. (2019), I exclude the firms before 1975. I clean the data as standard in the literature. I drop the firms that are not operating in the US firms. I drop financial and utility companies with SIC codes between 6000-7000, and 4900-5000. I remove the firms with zero or negative sales, cogs, ppegt, intangible capital, xsga and employment. I further exclude too small firms having employees less than 10. I, then, drop the firms that do not have naics industry classification codes. I remove the firms at the bottom 1% of the variables to omit the outliers. I do not exclude top 1% because some of the giant firms such as Amazon, Apple exclude from the data. Therefore, I only trim the data at the bottom 1%.

In the production function estimation I need output, physical capital, intangible capital, labor, material and knowledge capital. I use sale(Sales/Turnover (Net)) variable in Compustat for output. For the physical capital I use ppegt(Property, Plant and Equipment - Total

(Gross)) variable. I use cogs(cost of goods sold) variable for the material. emp (Employees in the Compustat captures the labor in the production function. In the data section I explain how to construct the intangible capital. Knowledge capital is the capital stock constructed xrd (Research & Development Expense variables in Compustat. I deflate all the variables with corresponding industry specific deflators in the BEA. Table B.2 represents some summary statistics for the variables used in my estimations(sales, ppegt, intangible capital, emp) after deflating and cleaning the data.

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